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Declaration

I certify that this report does not incorporate without acknowledgement, any material previously submitted for a degree or diploma in any university, and to the best of my knowledge and belief, it does not contain any material previously published or written by another person, except where due reference is made in text.

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1.Nagios Monitoring Setup with CentOS Server and Fedora Client/s

1.1 Setting up Nagios Core 4.5.1 within a CentOS Server-Fedora Client Architecture

```
[root@csa-server ~]# yum update
```

```
libtasn1.x86_64 0:4.10-1.el7
libuser.x86_64 0:0.60-9.el7
linux-firmware.noarch 0:20200421-02.git78c0340.el7_9
lsscsi.x86_64 0:0.27-6.el7
lvm2.x86_64 7:2.02.107-6.el7_9.5
man-db.x86_64 0:2.6.3-11.el7
microcode_ctl.x86_64 2:2.1-73.20.el7_9
ncurses.x86_64 0:5.9-14.20130511.el7_4
ncurses-libs.x86_64 0:5.9-14.20130511.el7_4
nspr.x86_64 0:4.35-0-1.el7_9
nss-softokn.x86_64 0:3.90.0-6.el7_9
nss-sysinit.x86_64 0:3.90.0-2.el7_9
nss-util.x86_64 0:3.90.0-1.el7_9
openssh.x86_64 0:7.4p1-23.el7_9
openssh-server.x86_64 0:7.4p1-23.el7_9
p11-kit.x86_64 0:0.23.5-3.el7
pam.x86_64 0:1.1.8-23.el7
passwd.x86_64 0:0.79-6.el7
pinentry.x86_64 0:0.8.1-17.el7
plymouth-core-libs.x86_64 0:0.8.9-0.34.20140113.el7.centos
polkit.x86_64 0:0.112-26.el7_9.1
ppp.x86_64 0:2.4.5-34.el7_9
python.x86_64 0:2.7.5-94.el7_9
python-perf.x86_64 0:3.10.0-1160.114.2.el7
python-pyudev.noarch 0:0.15-9.el7
python-slipdbus.noarch 0:0.4.0-4.el7
readline.x86_64 0:6.2-11.el7
rpm-build-libs.x86_64 0:4.11.3-48.el7_9
rpm-python.x86_64 0:4.11.3-48.el7_9
sed.x86_64 0:4.2.2-7.el7
shadow-utils.x86_64 2:4.6-5.el7
sqlite.x86_64 0:3.7.17-8.el7_9.1
tar.x86_64 2:1.26-35.el7
trousers.x86_64 0:0.3.14-2.el7
tzdata.noarch 0:2024a-1.el7
virt-what.x86_64 0:1.10-4.el7_9.1
xfsprogs.x86_64 0:4.5.0-22.el7
xz-libs.x86_64 0:5.2.2-2.el7_9
yum-plugin-fastestmirror.noarch 0:1.1.31-54.el7_0

Replaced:
NetworkManager.x86_64 1:1.0.6-29.el7_2
iwl7265-firmware.noarch 0:22.0.7.0-43.el7

grub2.x86_64 1:2.02-0.34.el7.centos
pygobject3-base.x86_64 0:3.14.0-3.el7

grub2-tools.x86_64 1:2.02-0.34.el7.centos
rdma.noarch 0:7.2.4.1_rc6-2.el7

Complete!
[root@csa-server ~]#
[root@csa-server ~]#
[root@csa-server ~]#
```

Figure 1.1. 1 : yum update

```
[root@csa-server ~]# yum install -y httpd httpd-tools php gcc glibc-common gd-devel make net-snmp_
```

Figure 1.1. 2 : install necessary packages

```
[root@csa-server ~]# yum install -y httpd httpd-tools php gcc glibc glibc-common gd gd-devel make net-snmp_
```

Figure 1.1. 3 : Install GCC C compiler with GLIBC libraries

```

[root@csa-server ~]# useradd nagios
[root@csa-server ~]# groupadd nagcmd
[root@csa-server ~]# _

```

```

[root@csa-server ~]# usermod -aG nagcmd nagios
[root@csa-server ~]# usermod -aG nagcmd apache
[root@csa-server ~]# _

```

Figure 1.1. 4 : Create Nagios user and group. Add user and Apache to the group

```

[root@csa-server ~]# cd nagios
[root@csa-server nagios]# wget https://assets.nagios.com/downloads/nagioscore/releases/nagios-4.5.1.tar.gz
--2024-04-21 14:19:51-- https://assets.nagios.com/downloads/nagioscore/releases/nagios-4.5.1.tar.gz
Resolving assets.nagios.com (assets.nagios.com)... 45.79.49.120, 2600:3c00::f03c:92ff:fe7:45ce
Connecting to assets.nagios.com (assets.nagios.com)|45.79.49.120|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 11540195 (11M) [application/x-gzip]
Saving to: 'nagios-4.5.1.tar.gz'

100%[=====>] 11,540,195  1.17MB/s  in 16s

2024-04-21 14:20:09 (696 KB/s) - 'nagios-4.5.1.tar.gz' saved [11540195/11540195]

[root@csa-server nagios]#

```

Figure 1.1. 5 : Move into Nagios folder and download Nagios 4.5.1 source code

```

[root@csa-server nagios]# wget http://nagios-plugins.org/download/nagios-plugins-2.4.9.tar.gz
--2024-04-21 14:31:50-- http://nagios-plugins.org/download/nagios-plugins-2.4.9.tar.gz
Resolving nagios-plugins.org (nagios-plugins.org)... 45.56.123.251
Connecting to nagios-plugins.org (nagios-plugins.org)|45.56.123.251|:80... connected.
HTTP request sent, awaiting response... 200 OK
Length: 2754403 (2.6M) [application/x-gzip]
Saving to: 'nagios-plugins-2.4.9.tar.gz'

100%[=====>] 2,754,403  712KB/s  in 3.8s

2024-04-21 14:31:54 (712 KB/s) - 'nagios-plugins-2.4.9.tar.gz' saved [2754403/2754403]

[root@csa-server nagios]# ls
nagios-4.5.1.tar.gz  nagios-plugins-2.4.9.tar.gz
[root@csa-server nagios]#

```

Figure 1.1. 6 : Download Nagios Plugins 2.4.9

```

[root@csa-server nagios]# ls
nagios-4.5.1.tar.gz  nagios-plugins-2.4.9.tar.gz
[root@csa-server nagios]#

```

Figure 1.1. 7 : Contents of 'nagios' folder

```

[root@csa-server nagios-4.5.1]# tar -xvf nagios-4.5.1.tar.gz_

```

Figure 1.1. 8 : Extract Nagios 4.5.1 file

```
[root@csa-server nagios]# tar -xvf nagios-plugins-2.4.9.tar.gz_
```

Figure 1.1. 9 : Extract Nagios-plugins 2.4.9

```
[root@csa-server nagios]# cd nagios-4.5.1
[root@csa-server nagios-4.5.1]# ls
aclocal.m4      common          configure       docs            indent-all.sh lib            mkpackage      pkginfo.in     subst.in       t-tap
autoconf-macros config.guess    configure.ac    doxy.conf       indent.sh      LICENSE        module         README.md      t             update-version
base           config.log     contrib        functions       INSTALLING    Makefile       nagios.spec    sample-config tap          UPGRADING
cgi            config.status  CONTRIBUTING.md html            install-sh     Makefile.in   nagios.sysconfig startup        test         worker
Changelog      config.sub     debian         include         LEGAL         make-tarball  pkginfo       subst         THANKS       xdata
[root@csa-server nagios-4.5.1]#
```

Figure 1.1. 10 Contents of 'nagios-4.5.1' folder

```
[root@csa-server nagios]# cd nagios-plugins-2.4.9
[root@csa-server nagios-plugins-2.4.9]# ls
ABOUT-NLS      AUTHORS        config.h.in    COPYING        LEGAL          Makefile.in    NEWS           pkg           po           SUPPORT       tools
acinclude.m4    build-aux     config_test    FAQ            lib            mkinstalldirs  NPTest.pm     plugins      README       tap          tools
ACKNOWLEDGEMENTS ChangeLog     configure      gl             m4            nagios-plugins.spec NP-VERSION-GEN plugins-root  release      test.pl.in
aclocal.m4      CODING        configure.ac    INSTALL        Makefile.am   nagios-plugins.spec.in perlmods       plugins-scripts REQUIREMENTS THANKS
[root@csa-server nagios-plugins-2.4.9]#
```

Figure 1.1. 11 Contents of 'nagios-pugins 2.4.9' folder

```
[root@csa-server ~]# cd nagios
[root@csa-server nagios]# cd nagios-4.5.1
[root@csa-server nagios-4.5.1]# ./configure --with-command-group=nagcmd
```

Figure 1.1. 12 Check system for required software to build Nagios 4.5.1

```

config.status: creating xdata/Makefile
config.status: creating subst
config.status: creating pkginfo
config.status: creating startup/openrc-init
config.status: creating startup/default-init
config.status: creating startup/default-service
config.status: creating startup/upstart-init
config.status: creating t/Makefile
config.status: creating t-tap/Makefile
config.status: creating include/config.h
config.status: include/config.h is unchanged
config.status: creating lib/snprintf.h
config.status: lib/snprintf.h is unchanged
config.status: creating lib/iobroker.h
config.status: lib/iobroker.h is unchanged

Creating sample config files in sample-config/ ...

*** Configuration summary for nagios 4.5.1 2024-02-28 ***:

General Options:
-----
Nagios executable: nagios
Nagios user/group: nagios,nagios
Command user/group: nagios,nagcmd
Event Broker: yes
Install ${prefix}: /usr/local/nagios
Install ${includedir}: /usr/local/nagios/include/nagios
Lock file: /run/nagios.lock
Check result directory: /usr/local/nagios/var/spool/checkresults
Init directory: /lib/systemd/system
Apache conf.d directory: /etc/httpd/conf.d
Mail program: /sbin/sendmail
Host OS: linux-gnu
IOBroker Method: epoll

Web Interface Options:
-----
HTML URL: http://localhost/nagios/
CGI URL: http://localhost/nagios/cgi-bin/
Traceroute (used by WAP):

Review the options above for accuracy. If they look okay,
type 'make all' to compile the main program and CGIs.

[root@csa-server nagios-4.5.1]#

```

Figure 1.1. 13 Configuration summary for Nagios 4.5.1

```

[root@csa-server nagios-4.5.1]# make all
cd ./base && make
make[1]: Entering directory '/root/nagios/nagios-4.5.1/base'
gcc -Wall -I. -I. -I./lib -I./include -I./include -I. -g -O2 -I/usr/include/krb5 -DHAVE_CONFIG_H -DNSCORE -c -o nagios.o ./nagios.c
gcc -Wall -I. -I. -I./lib -I./include -I./include -I. -g -O2 -I/usr/include/krb5 -DHAVE_CONFIG_H -DNSCORE -c -o broker.o broker.c
gcc -Wall -I. -I. -I./lib -I./include -I./include -I. -g -O2 -I/usr/include/krb5 -DHAVE_CONFIG_H -DNSCORE -c -o nebmodes.o nebmodes.c
gcc -Wall -I. -I. -I./lib -I./include -I./include -I. -g -O2 -I/usr/include/krb5 -DHAVE_CONFIG_H -DNSCORE -c -o ../common/shared.o ../common/shared.c
gcc -Wall -I. -I. -I./lib -I./include -I./include -I. -g -O2 -I/usr/include/krb5 -DHAVE_CONFIG_H -DNSCORE -c -o query-handler.o query-handler.c
gcc -Wall -I. -I. -I./lib -I./include -I./include -I. -g -O2 -I/usr/include/krb5 -DHAVE_CONFIG_H -DNSCORE -c -o workers.o workers.c

```

Figure 1.1. 14 Compile Nagios 4.5.1 source code

```

[root@csa-server nagios-4.5.1]#
[root@csa-server nagios-4.5.1]# make install_

```

Figure 1.1. 15 Place compiled files into default system path

```

make[1]: Leaving directory `/root/nagios/nagios-4.5.1'
make install-basic
make[1]: Entering directory `/root/nagios/nagios-4.5.1'
/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/libexec
/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/var
/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/var/archives
/bin/install -c -m 775 -o nagios -g nagcmd -d /usr/local/nagios/var/spool/checkresults
chmod g+s /usr/local/nagios/var/spool/checkresults

*** Main program, CGIs and HTML files installed ***

You can continue with installing Nagios as follows (type 'make'
without any arguments for a list of all possible options):

    make install-init
    - This installs the init script in /lib/systemd/system

    make install-commandmode
    - This installs and configures permissions on the
      directory for holding the external command file

    make install-config
    - This installs sample config files in /usr/local/nagios/etc

make[1]: Leaving directory `/root/nagios/nagios-4.5.1'
[root@csa-server nagios-4.5.1]#

```

Figure 1.1. 16 Install sample configuration files

```

[root@csa-server nagios-4.5.1]# make install-init
/bin/install -c -m 755 -d -o root -g root /lib/systemd/system
/bin/install -c -m 755 -o root -g root startup/default-service /lib/systemd/system/nagios.service
[root@csa-server nagios-4.5.1]# _

```

Figure 1.1. 17 Install init script

```

[root@csa-server nagios-4.5.1]# make install-commandmode
/bin/install -c -m 775 -o nagios -g nagcmd -d /usr/local/nagios/var/rw
chmod g+s /usr/local/nagios/var/rw

*** External command directory configured ***

[root@csa-server nagios-4.5.1]# _

```

Figure 1.1. 18 Configure external command directory

```

[root@csa-server nagios-4.5.1]# make install-config
/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/etc
/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/etc/objects
/bin/install -c -b -m 664 -o nagios -g nagios sample-config/nagios.cfg /usr/local/nagios/etc/nagios.cfg
/bin/install -c -b -m 664 -o nagios -g nagios sample-config/cgi.cfg /usr/local/nagios/etc/cgi.cfg
/bin/install -c -b -m 660 -o nagios -g nagios sample-config/resource.cfg /usr/local/nagios/etc/resource.cfg
/bin/install -c -b -m 664 -o nagios -g nagios sample-config/template-object/templates.cfg /usr/local/nagios/etc/objects/templates.cfg
/bin/install -c -b -m 664 -o nagios -g nagios sample-config/template-object/commands.cfg /usr/local/nagios/etc/objects/commands.cfg
/bin/install -c -b -m 664 -o nagios -g nagios sample-config/template-object/contacts.cfg /usr/local/nagios/etc/objects/contacts.cfg
/bin/install -c -b -m 664 -o nagios -g nagios sample-config/template-object/timeperiods.cfg /usr/local/nagios/etc/objects/timeperiods.cfg
/bin/install -c -b -m 664 -o nagios -g nagios sample-config/template-object/localhost.cfg /usr/local/nagios/etc/objects/localhost.cfg
/bin/install -c -b -m 664 -o nagios -g nagios sample-config/template-object/windows.cfg /usr/local/nagios/etc/objects/windows.cfg
/bin/install -c -b -m 664 -o nagios -g nagios sample-config/template-object/printer.cfg /usr/local/nagios/etc/objects/printer.cfg
/bin/install -c -b -m 664 -o nagios -g nagios sample-config/template-object/switch.cfg /usr/local/nagios/etc/objects/switch.cfg

*** Config files installed ***

Remember, these are *SAMPLE* config files. You'll need to read
the documentation for more information on how to actually define
services, hosts, etc. to fit your particular needs.

[root@csa-server nagios-4.5.1]#

```

Figure 1.1. 19 install sample configuration files

```
[root@csa-server nagios-4.5.1]# make install-webconf
/bin/install -c -m 644 sample-config/httpd.conf /etc/httpd/conf.d/nagios.conf
if [ 0 -eq 1 ]; then \
    ln -s /etc/httpd/conf.d/nagios.conf /etc/apache2/sites-enabled/nagios.conf; \
fi

*** Nagios/Apache conf file installed ***

[root@csa-server nagios-4.5.1]# _
```

Figure 1.1. 20 Install Nagios/Apache configuration file

```
[root@csa-server nagios-4.5.1]# sudo vi /usr/local/nagios/etc/objects/contacts.cfg_
```

Figure 1.1. 21 Modify 'contacts.cfg' file

```
# NOTES: This config file provides you with some example contact and contact
# group definitions that you can reference in host and service
# definitions.
#
# You don't need to keep these definitions in a separate file from your
# other object definitions. This has been done just to make things
# easier to understand.
#
#####

#####
#
# CONTACTS
#
#####

# Just one contact defined by default - the Nagios admin (that's you)
# This contact definition inherits a lot of default values from the
# 'generic-contact' template which is defined elsewhere.

define contact {
    contact_name      nagiosadmin          ; Short name of user
    use                generic-contact      ; Inherit default values from generic-contact template (defined above)
    alias              Nagios Admin         ; Full name of user
    email              it22014122@my.sliit.lk ; <<***** CHANGE THIS TO YOUR EMAIL ADDRESS *****>>
}

#####
#
# CONTACT GROUPS
#
#####

# We only have one contact in this simple configuration file, so there is
# no need to create more than one contact group.

define contactgroup {
    contactgroup_name admins
    alias              Nagios Administrators
    members             nagiosadmin
}
-- INSERT --
```

Figure 1.1. 22 Change email address of Nagios admin

```
[root@csa-server nagios-4.5.11]# sudo htpasswd -s -c /usr/local/nagios/etc/htpasswd.users nagiosadmin
New password:
Re-type new password:
Adding password for user nagiosadmin
[root@csa-server nagios-4.5.11]# _
```

Figure 1.1. 23 Create password for Nagios admin

```
[root@csa-server ~]# cd nagios
[root@csa-server nagios]# cd nagios-plugins-2.4.9
[root@csa-server nagios-plugins-2.4.9]# ./configure --with-nagios-user=nagios --with-nagios-group=nagcmd
```

Figure 1.1. 24 Check system for required software to build Nagios Plugins 2.4.9

```
config.status: creating po/Makefile
[root@csa-server nagios-plugins-2.4.9]# make _
```

Figure 1.1. 25 Compile Nagios Plugins 2.4.9 source code

```
[root@csa-server nagios-plugins-2.4.9]# make install
```

Figure 1.1. 26 Place compiled files into default system path

```
[root@csa-server nagios-plugins-2.4.9]# cd
[root@csa-server ~]# sudo /usr/local/nagios/bin/nagios -v /usr/local/nagios/etc/nagios.cfg _
```

Figure 1.1. 27 Verify configuration

```

make[1]: Leaving directory '/root/nagios/nagios-plugins-2.4.9/po'
make[1]: Entering directory '/root/nagios/nagios-plugins-2.4.9'
make[2]: Entering directory '/root/nagios/nagios-plugins-2.4.9'
make[2]: Nothing to be done for 'install-exec-am'.
make[2]: Nothing to be done for 'install-data-am'.
make[2]: Leaving directory '/root/nagios/nagios-plugins-2.4.9'
make[1]: Leaving directory '/root/nagios/nagios-plugins-2.4.9'
[root@csa-server nagios-plugins-2.4.9]# cd
[root@csa-server ~]# sudo /usr/local/nagios/bin/nagios -v /usr/local/nagios/etc/nagios.cfg

Nagios Core 4.5.1
Copyright (c) 2009-present Nagios Core Development Team and Community Contributors
Copyright (c) 1999-2009 Ethan Galstad
Last Modified: 2024-02-28
License: GPL

Website: https://www.nagios.org
Reading configuration data...
  Read main config file okay...
  Read object config files okay...

Running pre-flight check on configuration data...

Checking objects...
  Checked 8 services.
  Checked 1 hosts.
  Checked 1 host groups.
  Checked 0 service groups.
  Checked 1 contacts.
  Checked 1 contact groups.
  Checked 24 commands.
  Checked 5 time periods.
  Checked 0 host escalations.
  Checked 0 service escalations.
Checking for circular paths...
  Checked 1 hosts
  Checked 0 service dependencies
  Checked 0 host dependencies
  Checked 5 timeperiods
Checking global event handlers...
Checking obsessive compulsive processor commands...
Checking misc settings...

Total Warnings: 0
Total Errors: 0

Things look okay - No serious problems were detected during the pre-flight check
[root@csa-server ~]# _

```

Figure 1.1. 28 Configuration verified

```

[root@csa-server ~]# systemctl status httpd
■ httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; vendor preset: disabled)
   Active: active (running) since Wed 2024-04-24 04:44:54 EDT; 2min 17s ago
     Docs: man:httpd(8)
           man:apachectl(8)
  Main PID: 921 (httpd)
    Status: "Total requests: 0; Current requests/sec: 0; Current traffic:  0 B/sec"
    CGroup: /system.slice/httpd.service
            └─ 921 /usr/sbin/httpd -DFOREGROUND
              1082 /usr/sbin/httpd -DFOREGROUND
              1083 /usr/sbin/httpd -DFOREGROUND
              1084 /usr/sbin/httpd -DFOREGROUND
              1085 /usr/sbin/httpd -DFOREGROUND
              1086 /usr/sbin/httpd -DFOREGROUND

Apr 24 04:44:53 csa-server.csa.1k systemd[1]: Starting The Apache HTTP Server...
Apr 24 04:44:54 csa-server.csa.1k systemd[1]: Started The Apache HTTP Server.
[root@csa-server ~]# _

```

```

Things look okay - No serious problems were detected during the pre-flight check
[root@csa-server ~]# systemctl start nagios.service
[root@csa-server ~]# systemctl enable nagios.service
Created symlink from /etc/systemd/system/multi-user.target.wants/nagios.service to /usr/lib/systemd/system/nagios.service.
[root@csa-server ~]# systemctl status nagios.service
■ nagios.service - Nagios Core 4.5.1
   Loaded: loaded (/usr/lib/systemd/system/nagios.service; enabled; vendor preset: disabled)
   Active: active (running) since Tue 2024-04-23 10:35:13 EDT; 56s ago
     Docs: https://www.nagios.org/documentation
   Main PID: 94375 (nagios)
   CGroup: /system.slice/nagios.service
           └─94375 /usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nagios.cfg
             └─94376 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
               └─94377 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
                 └─94378 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
                   └─94379 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
                     └─94381 /usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nagios.cfg

Apr 23 10:35:13 csa-server.csa.lk nagios[94375]: qh: Socket '/usr/local/nagios/var/rw/nagios.qh' successfully initialized
Apr 23 10:35:13 csa-server.csa.lk nagios[94375]: qh: core query handler registered
Apr 23 10:35:13 csa-server.csa.lk nagios[94375]: qh: echo service query handler registered
Apr 23 10:35:13 csa-server.csa.lk nagios[94375]: qh: help for the query handler registered
Apr 23 10:35:13 csa-server.csa.lk nagios[94375]: wproc: Successfully registered manager as @wproc with query handler
Apr 23 10:35:13 csa-server.csa.lk nagios[94375]: wproc: Registry request: name=Core Worker 94377:pid=94377
Apr 23 10:35:13 csa-server.csa.lk nagios[94375]: wproc: Registry request: name=Core Worker 94379:pid=94379
Apr 23 10:35:13 csa-server.csa.lk nagios[94375]: wproc: Registry request: name=Core Worker 94376:pid=94376
Apr 23 10:35:13 csa-server.csa.lk nagios[94375]: wproc: Registry request: name=Core Worker 94378:pid=94378
Apr 23 10:35:13 csa-server.csa.lk nagios[94375]: Successfully launched command file worker with pid 94381
[root@csa-server ~]# _

```

Figure 1.1. 29 Nagios status

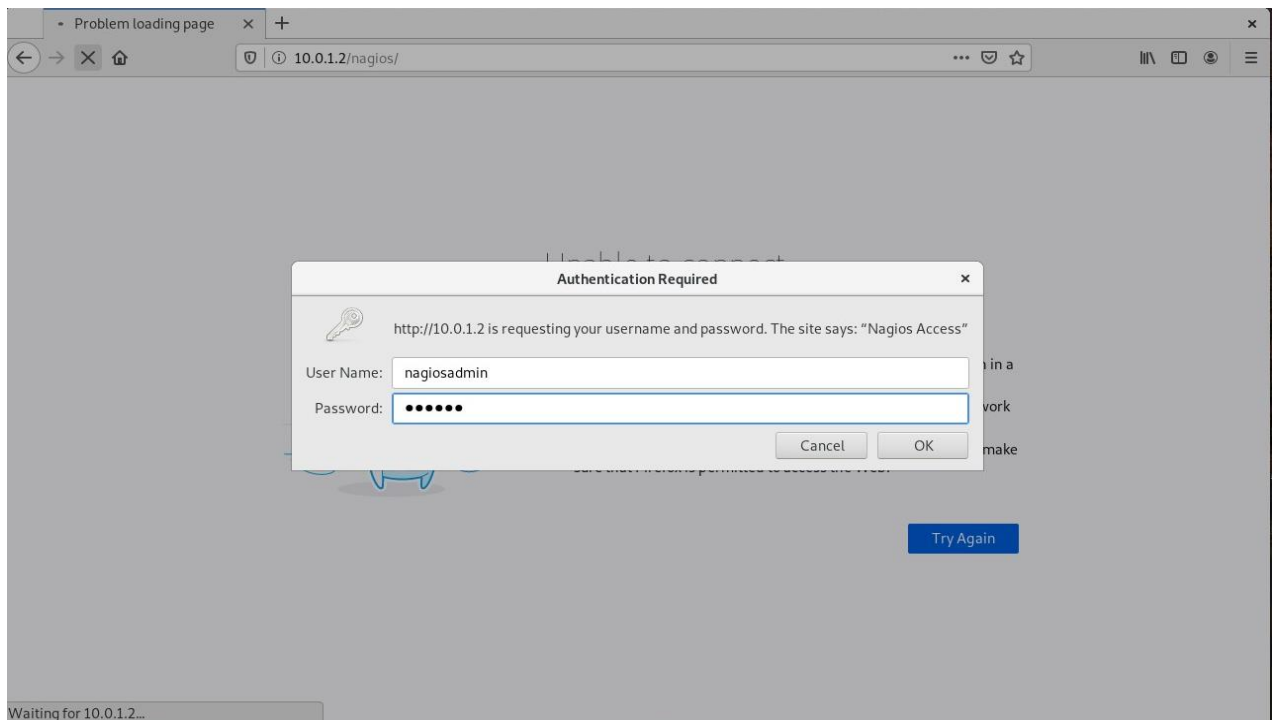


Figure 1.1. 30 Sign in as Nagios admin

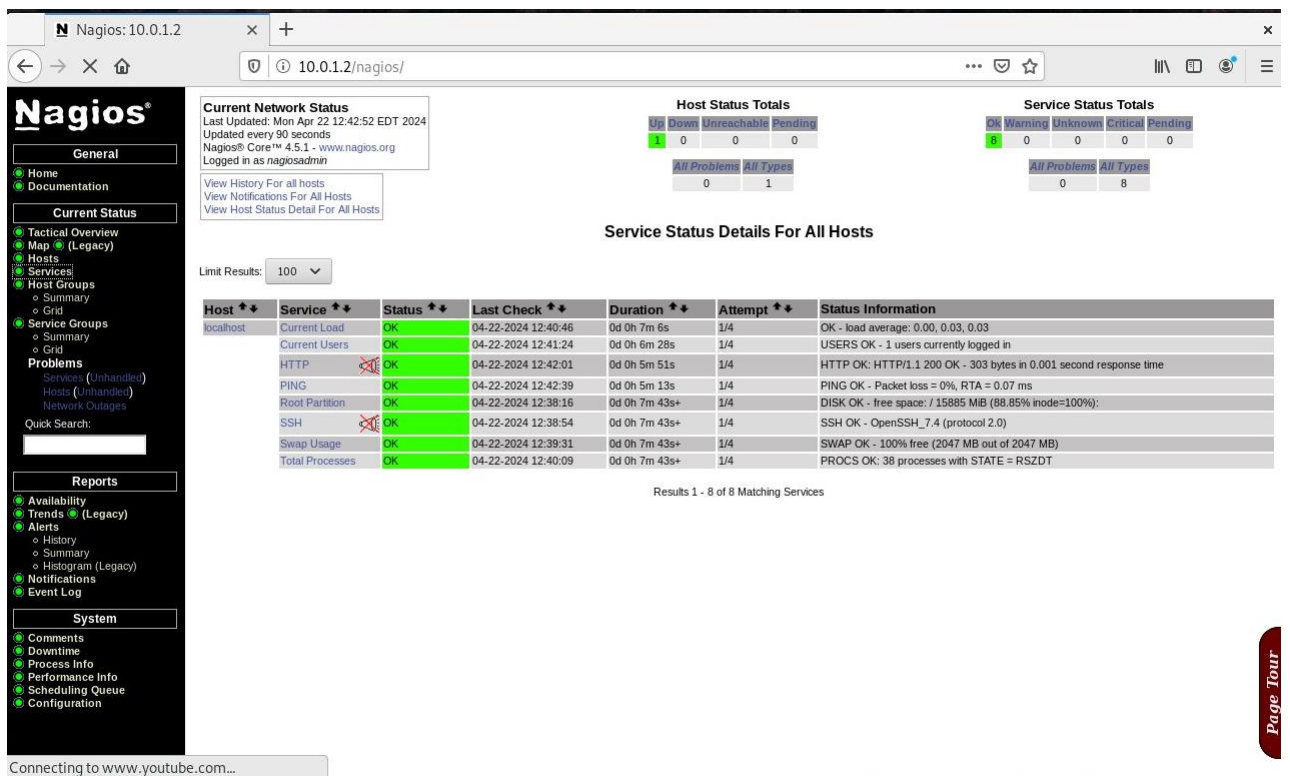


Figure 1.1. 31 Service status details for Nagios server

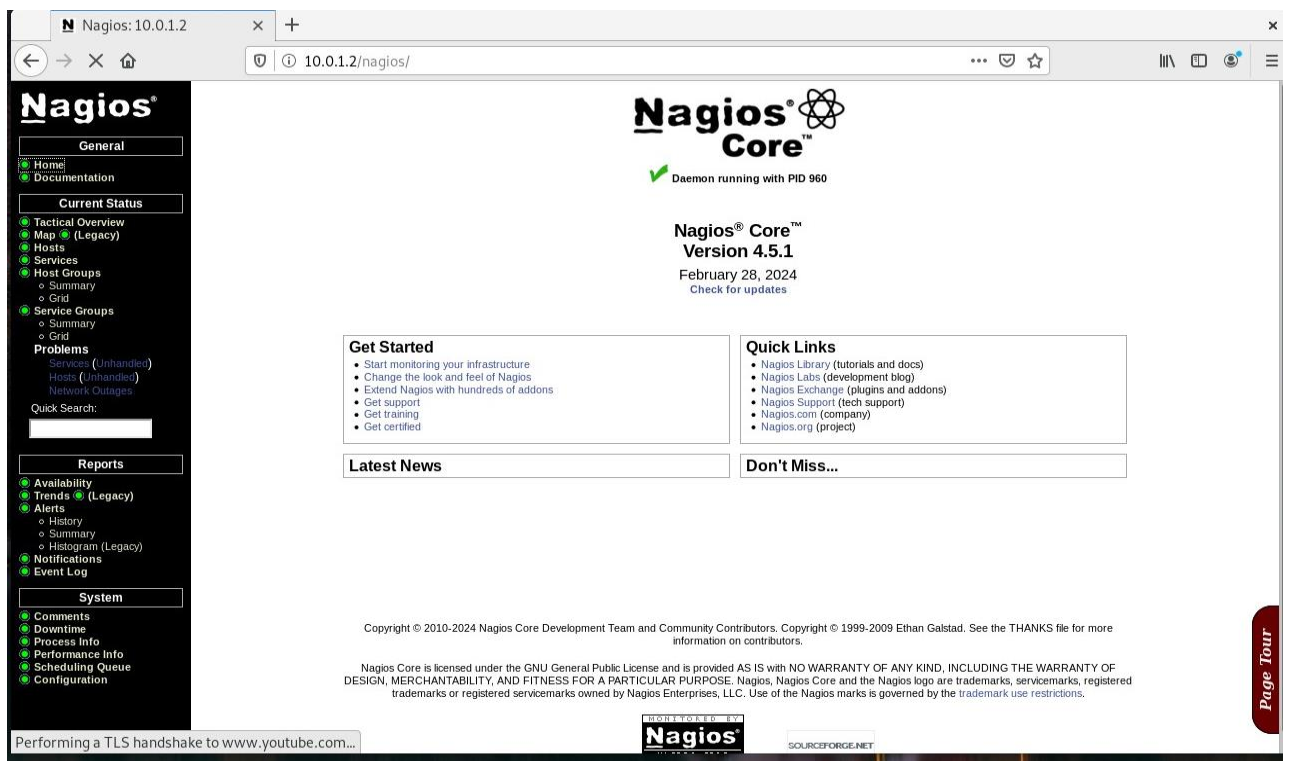


Figure 1.1. 32 Nagios Core homepage

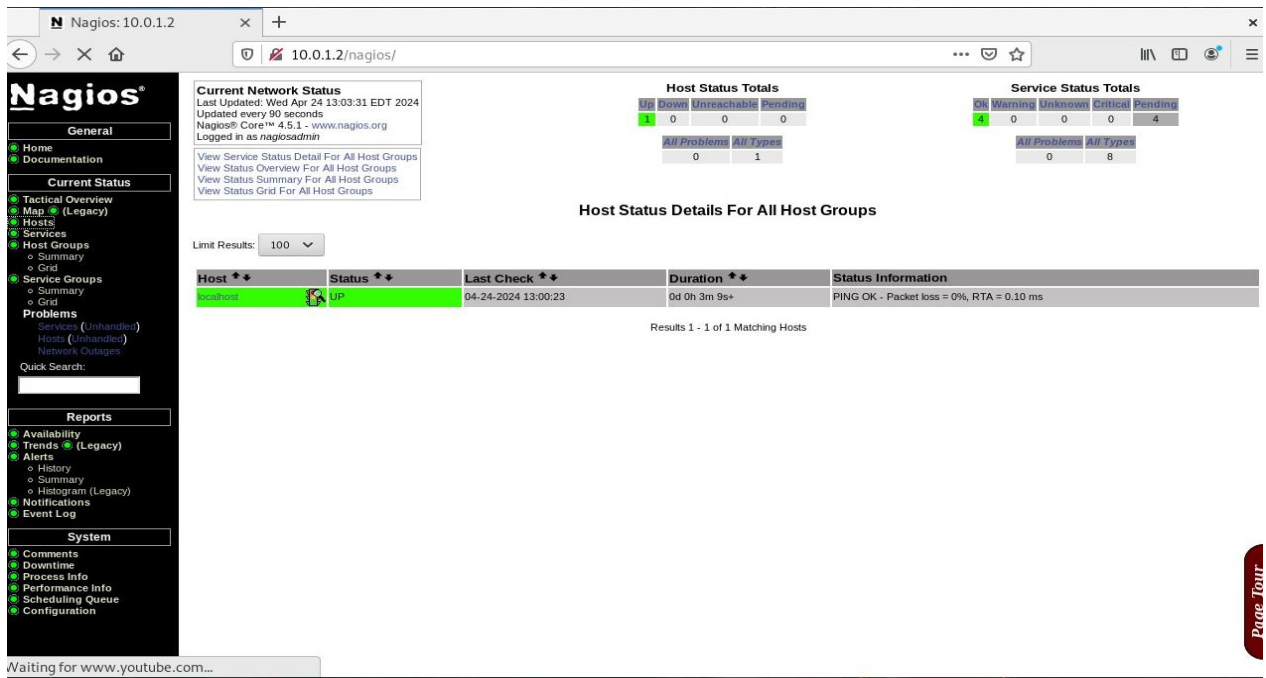


Figure 1.1. 33 Host status details for Nagios server

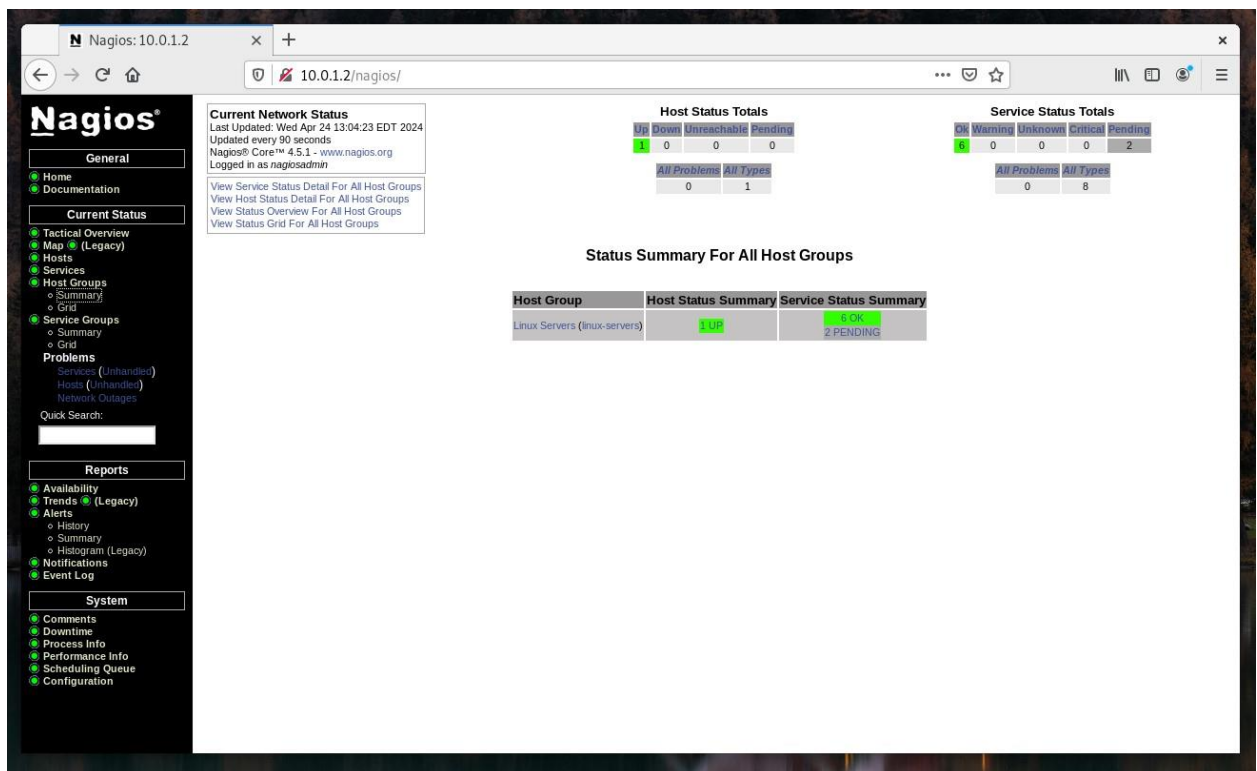
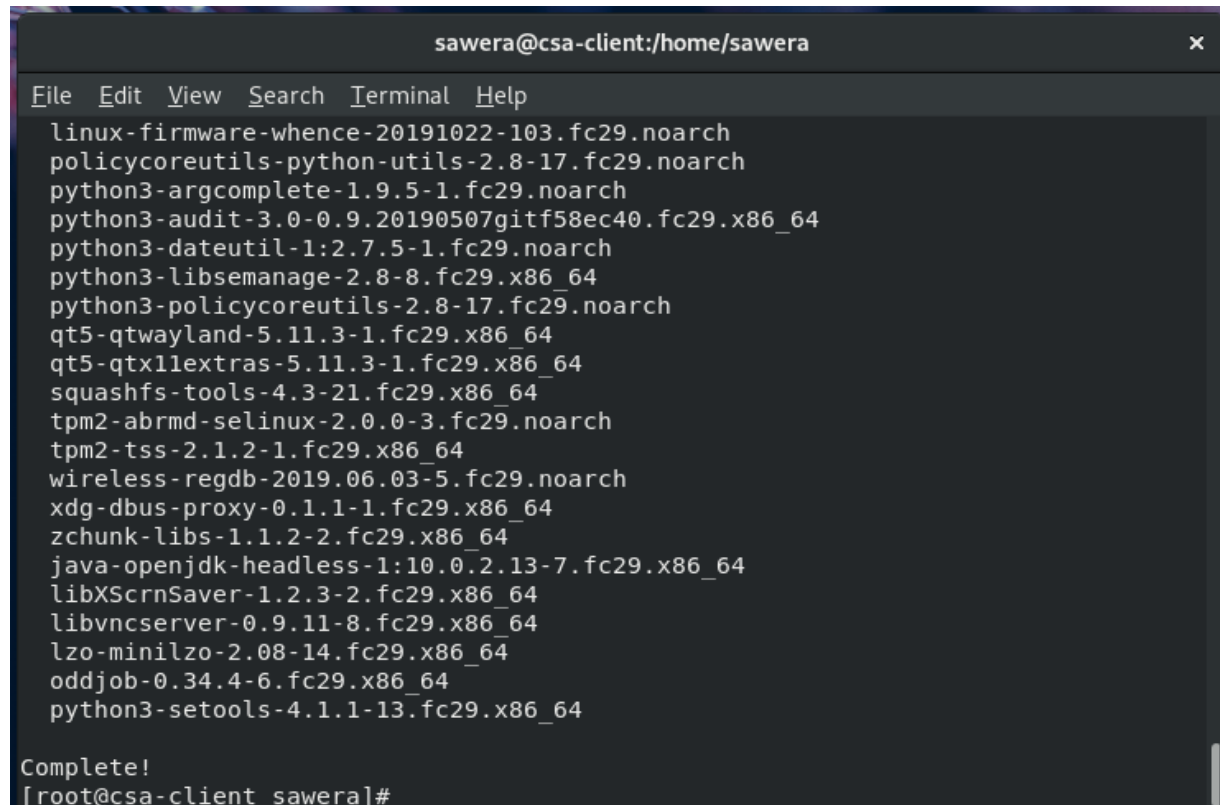


Figure 1.1. 34 Status summary for Nagios server host group

1.2 Configuring Nagios Server for Monitoring Fedora Client

```
[root@csa-client sawera]# dnf update
Last metadata expiration check: 0:09:36 ago on Fri 26 Apr 2024 04:13:14 PM +0530
.
Dependencies resolved.
=====
Package                                Arch    Version                                Repo    Size
=====
Upgrading:
java-openjdk-headless                  x86_64  1:12.0.0.33-4.rolling.fc29            updates 20 k
Installing dependencies:
java-1.8.0-openjdk-headless            x86_64  1:1.8.0.232.b09-0.fc29                updates 32 M
java-latest-openjdk-headless            x86_64  1:13.0.1.9-2.rolling.fc29             updates 41 M
Transaction Summary
=====
Install 2 Packages
Upgrade 1 Package

Total download size: 73 M
Is this ok [y/N]:
```



The image shows a terminal window titled 'sawera@csa-client:/home/sawera'. The terminal displays the output of a 'dnf install' command, listing various packages that were installed. The packages are listed in a single column, each on a new line. The list includes: linux-firmware-whence-20191022-103.fc29.noarch, polycoreutils-python-utils-2.8-17.fc29.noarch, python3-argcomplete-1.9.5-1.fc29.noarch, python3-audit-3.0-0.9.20190507gitf58ec40.fc29.x86_64, python3-dateutil-1:2.7.5-1.fc29.noarch, python3-libsemanage-2.8-8.fc29.x86_64, python3-polycoreutils-2.8-17.fc29.noarch, qt5-qtwayland-5.11.3-1.fc29.x86_64, qt5-qtquickcontrols-5.11.3-1.fc29.x86_64, squashfs-tools-4.3-21.fc29.x86_64, tpm2-abrmd-selinux-2.0.0-3.fc29.noarch, tpm2-tss-2.1.2-1.fc29.x86_64, wireless-regdb-2019.06.03-5.fc29.noarch, xdg-dbus-proxy-0.1.1-1.fc29.x86_64, zchunk-libs-1.1.2-2.fc29.x86_64, java-openjdk-headless-1:10.0.2.13-7.fc29.x86_64, libXScrnSaver-1.2.3-2.fc29.x86_64, libvncserver-0.9.11-8.fc29.x86_64, lzo-minilzo-2.08-14.fc29.x86_64, oddjob-0.34.4-6.fc29.x86_64, and python3-setuptools-4.1.1-13.fc29.x86_64. Below the list, the word 'Complete!' is displayed, followed by the prompt '[root@csa-client sawera]#'. The terminal window has a standard Linux desktop environment background and a title bar with the window name and a close button.

```
sawera@csa-client:/home/sawera
File Edit View Search Terminal Help
linux-firmware-whence-20191022-103.fc29.noarch
polycoreutils-python-utils-2.8-17.fc29.noarch
python3-argcomplete-1.9.5-1.fc29.noarch
python3-audit-3.0-0.9.20190507gitf58ec40.fc29.x86_64
python3-dateutil-1:2.7.5-1.fc29.noarch
python3-libsemanage-2.8-8.fc29.x86_64
python3-polycoreutils-2.8-17.fc29.noarch
qt5-qtwayland-5.11.3-1.fc29.x86_64
qt5-qtquickcontrols-5.11.3-1.fc29.x86_64
squashfs-tools-4.3-21.fc29.x86_64
tpm2-abrmd-selinux-2.0.0-3.fc29.noarch
tpm2-tss-2.1.2-1.fc29.x86_64
wireless-regdb-2019.06.03-5.fc29.noarch
xdg-dbus-proxy-0.1.1-1.fc29.x86_64
zchunk-libs-1.1.2-2.fc29.x86_64
java-openjdk-headless-1:10.0.2.13-7.fc29.x86_64
libXScrnSaver-1.2.3-2.fc29.x86_64
libvncserver-0.9.11-8.fc29.x86_64
lzo-minilzo-2.08-14.fc29.x86_64
oddjob-0.34.4-6.fc29.x86_64
python3-setuptools-4.1.1-13.fc29.x86_64
Complete!
[root@csa-client sawera]#
```

Figure1.2 1 dnf install and completed

```
[sawera@csa-client ~]$ sudo yum install -y gcc glibc glibc-common gd gd-devel make net-snmp openssl-devel tar wget
```

Figure1.2 2 Install necessary packages

```
[sawera@csa-client ~]$ sudo useradd nagios
[sawera@csa-client ~]$ sudo passwd nagios
Changing password for user nagios.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
[sawera@csa-client ~]$
```

Figure1.2 3 Add user and create password

```
[sawera@csa-client ~]$ id nagios
uid=1001(nagios) gid=1001(nagios) groups=1001(nagios)
[sawera@csa-client ~]$
```

Figure1.2 4 ID of the nagios user

```
[sawera@csa-client ~]$ sudo su
[sudo] password for sawera:
[root@csa-client sawera]# sudo dnf install xinetd
```

Figure1.2 5 Install xinetd

```
[root@csa-client ~]# mkdir nagios
[root@csa-client ~]# wget https://nagios-plugins.org/download/nagios-plugins-2.4.9.tar.gz
--2024-04-24 09:24:52-- https://nagios-plugins.org/download/nagios-plugins-2.4.9.tar.gz
Resolving nagios-plugins.org (nagios-plugins.org)... 45.56.123.251
Connecting to nagios-plugins.org (nagios-plugins.org)|45.56.123.251|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 2754403 (2.6M) [application/x-gzip]
Saving to: 'nagios-plugins-2.4.9.tar.gz'

nagios-plugins-2.4.9.tar.g 100%[=====] 2.63M 585KB/s in 5.1s

2024-04-24 09:25:00 (530 KB/s) - 'nagios-plugins-2.4.9.tar.gz' saved [2754403/2754403]

[root@csa-client ~]# ls
anaconda-ks.cfg nagios nagios-plugins-2.4.9.tar.gz
[root@csa-client ~]#
```

Figure1.2 6 Download Nagios-plugins 2.4.9

```

[root@csa-client nagios]# tar -xvf nagios-plugins-2.4.9.tar.gz
nagios-plugins-2.4.9/
nagios-plugins-2.4.9/build-aux/
nagios-plugins-2.4.9/build-aux/compile
nagios-plugins-2.4.9/build-aux/config.guess
nagios-plugins-2.4.9/build-aux/config.rpath
nagios-plugins-2.4.9/build-aux/config.sub
nagios-plugins-2.4.9/build-aux/install-sh
nagios-plugins-2.4.9/build-aux/ltmain.sh
nagios-plugins-2.4.9/build-aux/missing
nagios-plugins-2.4.9/build-aux/mkinstalldirs
nagios-plugins-2.4.9/build-aux/depcomp
nagios-plugins-2.4.9/build-aux/snippet/
nagios-plugins-2.4.9/build-aux/snippet/_Noreturn.h
nagios-plugins-2.4.9/build-aux/snippet/arg-nonnull.h
nagios-plugins-2.4.9/build-aux/snippet/c++defs.h
nagios-plugins-2.4.9/build-aux/snippet/warn-on-use.h
nagios-plugins-2.4.9/build-aux/test-driver
nagios-plugins-2.4.9/config_test/
nagios-plugins-2.4.9/config_test/Makefile
nagios-plugins-2.4.9/config_test/run_tests
nagios-plugins-2.4.9/config_test/child_test.c
nagios-plugins-2.4.9/gl/
nagios-plugins-2.4.9/gl/m4/
nagios-plugins-2.4.9/gl/m4/00gnulib.m4
nagios-plugins-2.4.9/gl/m4/absolute-header.m4
nagios-plugins-2.4.9/gl/m4/alloca.m4
nagios-plugins-2.4.9/gl/m4/arpa_inet_h.m4
nagios-plugins-2.4.9/gl/m4/base64.m4
nagios-plugins-2.4.9/gl/m4/btowc.m4

```

Figure1.2 7 Extract Nagios Plugins 2.4.9 file

```

[root@csa-client ~]# cd nagios-plugins-2.4.9/
[root@csa-client nagios-plugins-2.4.9]# ./configure
checking for a BSD-compatible install... /usr/bin/install -c
checking whether build environment is sane... yes
checking for a thread-safe mkdir -p... /usr/bin/mkdir -p
checking for gawk... gawk
checking whether make sets $(MAKE)... yes
checking whether make supports nested variables... yes
checking whether to enable maintainer-specific portions of Makefiles... yes
checking build system type... x86_64-pc-linux-gnu
checking host system type... x86_64-pc-linux-gnu

```

Figure1.2 8 Install software from source

```
[root@csa-client nagios-plugins-2.4.9]# make all
make all-recursive
make[1]: Entering directory '/root/nagios-plugins-2.4.9'
Making all in gl
make[2]: Entering directory '/root/nagios-plugins-2.4.9/gl'
rm -f alloca.h-t alloca.h && \
{ echo '/* DO NOT EDIT! GENERATED AUTOMATICALLY! */'; \
  cat ./alloca.in.h; \
} > alloca.h-t && \
mv -f alloca.h-t alloca.h
rm -f c++defs.h-t c++defs.h && \
sed -n -e '/_GL_CXXDEFS/, $p' \
  < ../build-aux/snippet/c++defs.h \
  > c++defs.h-t && \
mv c++defs.h-t c++defs.h
rm -f warn-on-use.h-t warn-on-use.h && \
sed -n -e '/^\.ifndef/, $p' \
  < ../build-aux/snippet/warn-on-use.h \
  > warn-on-use.h-t && \
mv warn-on-use.h-t warn-on-use.h
rm -f arg-nonnull.h-t arg-nonnull.h && \
sed -n -e '/GL_ARG_NONNULL/, $p' \
  < ../build-aux/snippet/arg-nonnull.h \
  > arg-nonnull.h-t && \
```

Figure1.2 9 Compile Nagios Plugins 2.4.9 source code

```
[root@csa-client nagios-plugins-2.4.9]# make install
Making install in gl
make[1]: Entering directory '/root/nagios-plugins-2.4.9/gl'
make install-recursive
make[2]: Entering directory '/root/nagios-plugins-2.4.9/gl'
make[3]: Entering directory '/root/nagios-plugins-2.4.9/gl'
make[4]: Entering directory '/root/nagios-plugins-2.4.9/gl'
if test yes = no; then \
  case 'linux-gnu' in \
    darwin[56]*) \
      need_charset_alias=true ;; \
    darwin* | cygwin* | mingw* | pw32* | cegcc*) \
      need_charset_alias=false ;; \
    *) \
      need_charset_alias=true ;; \
  esac ; \
else \
  need_charset_alias=false ; \
fi ; \
if $need_charset_alias; then \
  /bin/sh ../build-aux/mkinstalldirs /usr/local/nagios/lib ; \
fi ; \
if test -f /usr/local/nagios/lib/charset.alias; then \
  sed -f ref-add.sed /usr/local/nagios/lib/charset.alias > /usr/local/nagios/lib/charset.tmp ; \
  /usr/bin/install -c -m 644 /usr/local/nagios/lib/charset.tmp /usr/local/nagios/lib/charset.alias ; \
```

Figure1.2 10 Place compiled files into default system path

```
[root@csa-client ~]# chown nagios:nagios /usr/local/nagios
[root@csa-client ~]# ls -l /usr/local/nagios
total 8
drwxr-xr-x. 2 root root 4096 Apr 24 11:12 libexec
drwxr-xr-x. 3 root root 4096 Apr 24 11:12 share
[root@csa-client ~]# chown -R nagios:nagios /usr/local/nagios/libexec
[root@csa-client ~]# ls -l /usr/local/nagios
total 8
drwxr-xr-x. 2 nagios nagios 4096 Apr 24 11:12 libexec
drwxr-xr-x. 3 root root 4096 Apr 24 11:12 share
```

Figure1.2 11 List the contents of the Nagios folder

```
[root@csa-client ~]# cd nagios
[root@csa-client nagios]# wget https://github.com/NagiosEnterprises/nrpe/releases/download/nrpe-4.1.0/nrpe-4.1.0.tar.gz
--2024-04-24 11:39:24-- https://github.com/NagiosEnterprises/nrpe/releases/download/nrpe-4.1.0/nrpe-4.1.0.tar.gz
Resolving github.com (github.com)... 20.205.243.166
Connecting to github.com (github.com)|20.205.243.166|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://objects.githubusercontent.com/github-production-release-asset-2e65be/16119653/15952b1c-13e6-4e48-939d-292348427f8c?X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Credential=AKIAVCODYLSA53PQK4ZA%2F20240424%2Fus-east-1%2Fs3%2Faws4_request&X-Amz-Date=20240424T060844Z&X-Amz-Expires=300&X-Amz-Signature=5f69f113f920bc2741d3fa1034de1b4efe496688d546b1e38938b8f254c0c8526&X-Amz-SignedHeaders=host&actor_id=0&key_id=0&repo_id=16119653&response-content-disposition=attachment%3B%20filename%3Dnrpe-4.1.0.tar.gz&response-content-type=application%2Foctet-stream [following]
--2024-04-24 11:39:25-- https://objects.githubusercontent.com/github-production-release-asset-2e65be/16119653/15952b1c-13e6-4e48-939d-292348427f8c?X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Credential=AKIAVCODYLSA53PQK4ZA%2F20240424%2Fus-east-1%2Fs3%2Faws4_request&X-Amz-Date=20240424T060844Z&X-Amz-Expires=300&X-Amz-Signature=5f69f113f920bc2741d3fa1034de1b4efe496688d546b1e38938b8f254c0c8526&X-Amz-SignedHeaders=host&actor_id=0&key_id=0&repo_id=16119653&response-content-disposition=attachment%3B%20filename%3Dnrpe-4.1.0.tar.gz&response-content-type=application%2Foctet-stream
Resolving objects.githubusercontent.com (objects.githubusercontent.com)... 185.199.108.133, 185.199.111.133, 185.199.110.133, ...
Connecting to objects.githubusercontent.com (objects.githubusercontent.com)|185.199.108.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 528228 (516K) [application/octet-stream]
Saving to: 'nrpe-4.1.0.tar.gz'

nrpe-4.1.0.tar.gz      100%[=====] 515.85K  257KB/s   in 2.0s

2024-04-24 11:39:29 (257 KB/s) - 'nrpe-4.1.0.tar.gz' saved [528228/528228]

[root@csa-client nagios]# ls
nrpe-4.1.0.tar.gz
[root@csa-client nagios]#
```

Figure1.2 12 Download Nagios Remote Plugin Executor (NRPE) 4.1.0

```

[root@csa-client nagios]# tar -xvf nrpe-4.1.0.tar.gz
nrpe-4.1.0/
nrpe-4.1.0/.gitignore
nrpe-4.1.0/.travis.yml
nrpe-4.1.0/CHANGELOG.md
nrpe-4.1.0/CONTRIBUTING.md
nrpe-4.1.0/LEGAL
nrpe-4.1.0/LICENSE.md
nrpe-4.1.0/Makefile.in
nrpe-4.1.0/README.SSL.md
nrpe-4.1.0/README.md
nrpe-4.1.0/SECURITY.md
nrpe-4.1.0/THANKS
nrpe-4.1.0/aclocal.m4
nrpe-4.1.0/build-aux/
nrpe-4.1.0/build-aux/config.guess
nrpe-4.1.0/build-aux/config.sub
nrpe-4.1.0/build-aux/custom_help.m4
nrpe-4.1.0/build-aux/install-sh
nrpe-4.1.0/configure
nrpe-4.1.0/configure.ac
nrpe-4.1.0/contrib/
nrpe-4.1.0/contrib/README.nrpe_check_control
nrpe-4.1.0/contrib/nrpe_check_control.c
nrpe-4.1.0/docs/
nrpe-4.1.0/docs/NRPE.pdf

```

Figure1.2 13 Extract Nagios nrpe 4.1.0 file

```

[root@csa-client nagios]# cd nrpe-4.1.0/
[root@csa-client nrpe-4.1.0]# ./configure

```

Figure1.2 14 heck system for required software to build Nagios nrpe 4.1.0

```

config.status: creating startup/default-xinetd
config.status: creating startup/mac-init.plist
config.status: creating startup/mac-inetd.plist
config.status: creating startup/newbsd-init
config.status: creating startup/openbsd-init
config.status: creating startup/openrc-conf
config.status: creating startup/openrc-init
config.status: creating startup/solaris-init.xml
config.status: creating startup/solaris-inetd.xml
config.status: creating startup/tmpfile.conf
config.status: creating startup/upstart-init
config.status: creating startup/rh-upstart-init
config.status: creating include/common.h
config.status: creating include/config.h

*** Configuration summary for nrpe 4.1.0 2022-07-18 ***:

General Options:
-----
NRPE port:      5666
NRPE user:      nagios
NRPE group:     nagios
Nagios user:    nagios
Nagios group:   nagios

Review the options above for accuracy.  If they look okay,
type 'make all' to compile the NRPE daemon and client
or type 'make' to get a list of make options.

```

Figure1.2 15 Make file generated

```
[root@csa-client nrpe-4.1.0]# make all
cd ./src/; make
make[1]: Entering directory '/root/nagios/nrpe-4.1.0/src'
gcc -g -O2 -DHAVE_CONFIG_H -I ../include -I ../../include -o nrpe ./nrpe.c ./utils.c ./acl.c -lssl -lcrypto
gcc -g -O2 -DHAVE_CONFIG_H -I ../include -I ../../include -o check_nrpe ./check_nrpe.c ./utils.c -lssl -lcrypto
make[1]: Leaving directory '/root/nagios/nrpe-4.1.0/src'

*** Compile finished ***

You can now continue with the installation or upgrade process.

Read the PDF documentation (docs/NRPE.pdf) for information on the next
steps you should take to complete the installation or upgrade.
```

Figure1.2 16 Compile NRPE source code

```
[root@csa-client nrpe-4.1.0]# make install
cd ./src/; make install
make[1]: Entering directory '/root/nagios/nrpe-4.1.0/src'
make install-plugin
make[2]: Entering directory '/root/nagios/nrpe-4.1.0/src'
/usr/bin/install -c -m 755 -d /usr/local/nagios/bin
/usr/bin/install -c -m 755 ../uninstall /usr/local/nagios/bin/nrpe-uninstall
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/libexec
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/libexec
/usr/bin/install -c -m 775 -o nagios -g nagios check_nrpe /usr/local/nagios/libexec
make[2]: Leaving directory '/root/nagios/nrpe-4.1.0/src'
make install-daemon
make[2]: Entering directory '/root/nagios/nrpe-4.1.0/src'
/usr/bin/install -c -m 755 -d /usr/local/nagios/bin
/usr/bin/install -c -m 755 ../uninstall /usr/local/nagios/bin/nrpe-uninstall
/usr/bin/install -c -m 755 nrpe /usr/local/nagios/bin
/usr/bin/install -c -m 755 -o nagios -g nagios -d /usr/local/nagios/var
/usr/bin/install -c -m 755 -d /usr/lib/tmpfiles.d
/usr/bin/install -c -m 644 ../startup/tmpfile.conf /usr/lib/tmpfiles.d/nrpe.conf
make[2]: Leaving directory '/root/nagios/nrpe-4.1.0/src'
make[1]: Leaving directory '/root/nagios/nrpe-4.1.0/src'
```

Figure1.2 17 Add user and place compiled files into default system path

```

[root@csa-client nrpe-4.1.0]# make install-daemon
cd ./src/; make install-daemon
make[1]: Entering directory '/root/nagios/nrpe-4.1.0/src'
/usr/bin/install -c -m 755 -d /usr/local/nagios/bin
/usr/bin/install -c -m 755 ../uninstall /usr/local/nagios/bin/nrpe-uninstall
/usr/bin/install -c -m 755 nrpe /usr/local/nagios/bin
/usr/bin/install -c -m 755 -d /usr/lib/tmpfiles.d
/usr/bin/install -c -m 644 ../startup/tmpfile.conf /usr/lib/tmpfiles.d/nrpe.conf
make[1]: Leaving directory '/root/nagios/nrpe-4.1.0/src'
[root@csa-client nrpe-4.1.0]# make install-config
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/etc
/usr/bin/install -c -m 644 -o nagios -g nagios sample-config/nrpe.cfg /usr/local/nagios/etc
[root@csa-client nrpe-4.1.0]# cd
[root@csa-client ~]# systemctl enable nrpe
Failed to enable unit: Unit file nrpe.service does not exist.
[root@csa-client ~]# dnf install nrpe
Last metadata expiration check: 0:08:09 ago on Wed 24 Apr 2024 12:57:31 PM +0530.
Dependencies resolved.
=====
Package                                Architecture      Version            Repository         Size
=====
Installing:
nrpe                                    x86_64            3.2.1-6.fc29      fedora             260 k
Installing dependencies:
nagios-common                          x86_64            4.4.3-1.fc29      updates            13 k
=====
Transaction Summary
=====
Install 2 Packages

Total download size: 273 k

```

```

[root@csa-client nrpe-4.1.0]# make install-plugin
cd ./src/; make install-plugin
make[1]: Entering directory '/root/nagios/nrpe-4.1.0/src'
/usr/bin/install -c -m 755 -d /usr/local/nagios/bin
/usr/bin/install -c -m 755 ../uninstall /usr/local/nagios/bin/nrpe-uninstall
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/libexec
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/libexec
/usr/bin/install -c -m 775 -o nagios -g nagios check_nrpe /usr/local/nagios/libexec
make[1]: Leaving directory '/root/nagios/nrpe-4.1.0/src'
[root@csa-client nrpe-4.1.0]# make install-daemon
cd ./src/; make install-daemon
make[1]: Entering directory '/root/nagios/nrpe-4.1.0/src'
/usr/bin/install -c -m 755 -d /usr/local/nagios/bin
/usr/bin/install -c -m 755 ../uninstall /usr/local/nagios/bin/nrpe-uninstall
/usr/bin/install -c -m 755 nrpe /usr/local/nagios/bin
/usr/bin/install -c -m 755 -d /usr/lib/tmpfiles.d
/usr/bin/install -c -m 644 ../startup/tmpfile.conf /usr/lib/tmpfiles.d/nrpe.conf
make[1]: Leaving directory '/root/nagios/nrpe-4.1.0/src'
[root@csa-client nrpe-4.1.0]# make install-config
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/etc
/usr/bin/install -c -m 644 -o nagios -g nagios sample-config/nrpe.cfg /usr/local/nagios/etc

```

Figure1.2 18 Compile and install plugins from source

```

[root@csa-client ~]# systemctl enable nrpe
Created symlink /etc/systemd/system/multi-user.target.wants/nrpe.service → /usr/lib/systemd/system/nrpe.service.
[root@csa-client ~]# systemctl start nrpe
[root@csa-client ~]# /usr/local/nagios/libexec/check
bash: /usr/local/nagios/libexec/check: No such file or directory
[root@csa-client ~]# /usr/local/nagios/libexec/check_nrpe -H 127.0.0.1
NRPE v3.2.1

```

Figure1.2 19 Test NRPE plugin

```

[root@csa-client nrpe-4.1.0]# id nagios
uid=1001(nagios) gid=1001(nagios) groups=1001(nagios)

```

Figure1.2 20 ID of user nagios

```

[root@csa-client ~]# ls
anaconda-ks.cfg  nagios-plugins-2.4.9  nrpe-4.1.0
nagios           nagios-plugins-2.4.9.tar.gz  nrpe-4.1.0.tar.gz
[root@csa-client ~]# cd nrpe-4.1.0
[root@csa-client nrpe-4.1.0]# ls
aclocal.m4      contrib      LICENSE.md    README.md     startup      update-cfg.pl
build-aux      CONTRIBUTING.md  macros      README.SSL.md  test-wrapper  update-version
CHANGELOG.md   docs         Makefile.in   sample-config  THANKS
configure      include      nrpe.spec.in  SECURITY.md    travis-test-1
configure.ac   LEGAL        paths.in      src            uninstall.in
[root@csa-client nrpe-4.1.0]#

```

Figure1.2 21 Contents of 'nrpe-4.1.0' folder

```

[root@csa-client sample-config]# vi nrpe.xinetd

```

Figure1.2 22 Create 'nrpe.xinetd' file

[illegible]

Figure1.2 23 ‘nrpe.xinetd’ file created

```
[root@csa-client ~]# cd /etc/xinetd.d
[root@csa-client xinetd.d]# ls
chargen-dgram  daytime-dgram  discard-dgram  echo-dgram  tcpmux-server  time-stream
chargen-stream  daytime-stream  discard-stream  echo-stream  time-dgram
[root@csa-client xinetd.d]# vi nrpe.xinetd
```

Figure 1.24 Copy file into directory containing xinetd configuration files

```
[root@csa-client sample-config]# vi nrpe.xinetd
[root@csa-client sample-config]# cd
[root@csa-client ~]# cd /etc/xinetd.d
```

Figure1.2 25 Edit the xinetd configuration file for NRPE service on port 5666

```
# default: on
# description: NRPE (Nagios Remote Plugin Executor)
service nrpe
{
    flags            = REUSE
    type            = UNLISTED
    port            = 5666
    socket_type      = stream
    wait            = no
    user            = nagios
    group           = nagios
    server           = /usr/sbin/nrpe
    server_args      = -c /etc/nagios/nrpe.cfg --inetd
    log_on_failure   += USERID
    disable         = no
    only_from        = 10.0.1.2
}

~
~
~
~
~

:wq!
```

Figure1.2 26 NRPE xinetd configuration file edited

```
[root@csa-client xinetd.d]# cd
[root@csa-client ~]# systemctl restart xinetd
[root@csa-client ~]#
```

Figure1.2 27 Restart xinetd service

```
[root@csa-client ~]# vi /etc/services
```

Figure1.2 28 Check the configuration file that maps port numbers to named services

```
xprtld      5634/udp      # SF Message Service
sfmsso      5635/tcp      # SFM Authentication Subsystem
sfm-db-server 5636/tcp      # SFMdb - SFM DB server
cssc        5637/tcp      # Symantec CSSC
flcrs       5638/tcp      # Symantec Fingerprint Lookup and Container Reference
lcs         5639/tcp      # Symantec Integrity Checking
vfmobile    5646/tcp      # Ventureforth Mobile
nrpe        5666/tcp      # Nagios Remote Plugin Executor
filemq      5670/tcp      # ZeroMQ file
zre-disc    5670/udp      # Local area discovery and msging over ZeroMQ
amqps       5671/tcp      # amqp protocol over TLS/SSL
amqps       5671/udp      # amqp protocol over TLS/SSL
amqp        5672/tcp      # AMQP
amqp        5672/udp      # AMQP
amqp        5672/sctp     # AMQP
jms         5673/tcp      # JACL Message Server
jms         5673/udp      # JACL Message Server
hyperscsi-port 5674/tcp      # HyperSCSI Port
hyperscsi-port 5674/udp      # HyperSCSI Port
v5ua        5675/tcp      # V5UA application port
v5ua        5675/udp      # V5UA application port
v5ua        5675/sctp     # V5UA application port
raadmin     5676/tcp      # RA Administration
```

Figure1.2 29 NRPE entry is present

```
[root@csa-client nagios]# cd nrpe-4.1.0
[root@csa-client nrpe-4.1.0]# cd sample-config/
[root@csa-client sample-config]# more nrpe.xinetd
# default: on
# description: NRPE (Nagios Remote Plugin Executor)
service nrpe
{
    flags            = REUSE
    type             = UNLISTED
    port            = 5666
    socket_type      = stream
    wait            = no
    user            = nagios
    group           = nagios
    server          = /usr/sbin/nrpe
    server_args      = -c /etc/nagios/nrpe.cfg --inetd
    log_on_failure   += USERID
    disable         = no
#    only_from       = 127.0.0.1
}
[root@csa-client sample-config]# █
```

Figure1.2 30 Check the path to be created for NRPE configuration file

```
[root@csa-client ~]# mkdir /usr/local/nagios/etc/█
```

Figure1.2 31 Path for NRPE configuration file created

```
[root@csa-client nagios]# cd nrpe-4.1.0
[root@csa-client nrpe-4.1.0]# cd sample-config
[root@csa-client sample-config]# cp nrpe.cfg /usr/local/nagios/etc
cp: overwrite '/usr/local/nagios/etc/nrpe.cfg'? no
[root@csa-client sample-config]# cd /usr/local/nagios/etc
[root@csa-client etc]# ls
nrpe.cfg
[root@csa-client etc]# █
```

Figure1.2 32 Copy NRPE configuration file to the path and edit the file

```
[root@csa-client etc]# vi nrpe.cfg█
```

```

# PORT NUMBER
# Port number we should wait for connections on.
# NOTE: This must be a non-privileged port (i.e. > 1024).
# NOTE: This option is ignored if NRPE is running under either inetd or xinetd

server_port=5666

# SERVER ADDRESS
# Address that nrpe should bind to in case there are more than one interface
# and you do not want nrpe to bind on all interfaces.
# NOTE: This option is ignored if NRPE is running under either inetd or xinetd

server_address=10.0.1.2

# LISTEN QUEUE SIZE
# Listen queue size (backlog) for serving incoming connections.
# You may want to increase this value under high load.
-- INSERT --

# Note: The daemon only does rudimentary checking of the client's IP
# address. I would highly recommend adding entries in your /etc/hosts.allow
# file to allow only the specified host to connect to the port
# you are running this daemon on.
#
# NOTE: This option is ignored if NRPE is running under either inetd or xinetd

allowed_hosts=127.0.0.1,10.0.1.2

# COMMAND ARGUMENT PROCESSING
# This option determines whether or not the NRPE daemon will allow clients
# to specify arguments to commands that are executed. This option only works
# if the daemon was configured with the --enable-command-args configure script
# option.
#
# *** ENABLING THIS OPTION IS A SECURITY RISK! ***
# Read the SECURITY file for information on some of the security implications
# of enabling this variable.
#
# Values: 0=do not allow arguments, 1=allow command arguments
█
-- INSERT --

```

Figure1.2 33 Change server address and allowed hosts in the file

```

[root@csa-client etc]# systemctl start nrpe
[root@csa-client etc]# systemctl enable nrpe
[root@csa-client etc]# systemctl restart nrpe
[root@csa-client etc]# █

```

Figure1.2 34 Start and enable NRPE

```

[root@csa-client etc]# cd /usr/local/nagios/libexec
[root@csa-client libexec]# pwd
/usr/local/nagios/libexec
[root@csa-client libexec]# ls
check_apr      check_dns      check_ifstatus  check_nagios    check_oracle    check_smtp      check_ups
check_breeze   check_dummy    check_imap      check_nntp      check_overcr    check_spop      check_uptime
check_by_ssh   check_file_age  check_ircd      check_nntp      check_ping      check_ssh       check_users
check_clamd    check_flexlm    check_jabber    check_nrpe      check_pop       check_ssl_validity  check_wave
check_cluster  check_ftp      check_load      check_nt        check_procs     check_ssmtp     negate
check_dhcp     check_http     check_log       check_ntp       check_real      check_swap      remove_perfdata
check_dig      check_icmp     check_mailq     check_ntp_peer  check_rpc       check_tcp       urlize
check_disk     check_ide_smart  check_mrtg     check_ntp_time  check_sensors   check_time      utils.pm
check_disk_smb  check_ifoperstatus  check_mrtgtraf  check_nwstat   check_simap     check_udp       utils.sh
[root@csa-client libexec]# ./check_nrpe -H 127.0.0.1
NRPE v3.2.1
[root@csa-client libexec]#

```

Figure1.2 35 Test NRPE plugin

```

Fedora Modular 29 - x86_64                17 kB/s | 6.4 kB    00:00
Fedora Modular 29 - x86_64 - Updates      29 kB/s | 6.3 kB    00:00
Fedora 29 - x86_64 - Updates              6.9 kB/s | 6.4 kB    00:00
Fedora 29 - x86_64                        7.0 kB/s | 6.5 kB    00:00
Dependencies resolved.
=====
Package                Architecture Version                Repository            Size
=====
Installing:
iptables-services      x86_64        1.8.0-3.fc29          fedora                15 k
Transaction Summary
=====
Install 1 Package

Total download size: 15 k
Installed size: 22 k
Is this ok [y/N]: y
Downloading Packages:
iptables-services-1.8.0-3.fc29.x86_64.rpm 15 kB/s | 15 kB    00:01
-----
Total                                     9.2 kB/s | 15 kB    00:01
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      :                                1/1
  Installing     : iptables-services-1.8.0-3.fc29.x86_64 1/1
  Running scriptlet: iptables-services-1.8.0-3.fc29.x86_64 1/1
  Verifying      : iptables-services-1.8.0-3.fc29.x86_64 1/1

Installed:
iptables-services-1.8.0-3.fc29.x86_64

Complete!
[root@csa-client deliya]# █

```

Figure1.2 36 Install Linux firewall


```
[root@csa-client sysconfig]# systemctl restart iptables
[root@csa-client sysconfig]# systemctl status iptables
● iptables.service - IPv4 firewall with iptables
   Loaded: loaded (/usr/lib/systemd/system/iptables.service; disabled; vendor preset: disabled)
   Active: active (exited) since Thu 2024-04-25 12:54:05 +0530; 23s ago
   Process: 2984 ExecStart=/usr/libexec/iptables/iptables.init start (code=exited, status=0/SUCCESS)
   Main PID: 2984 (code=exited, status=0/SUCCESS)

Apr 25 12:54:05 csa-client.csa.lk- systemd[1]: Starting IPv4 firewall with iptables...
Apr 25 12:54:05 csa-client.csa.lk- iptables.init[2984]: iptables: Applying firewall rules: [ OK ]
Apr 25 12:54:05 csa-client.csa.lk- systemd[1]: Started IPv4 firewall with iptables.
[root@csa-client sysconfig]#
```

Figure1.2 39 Restart the firewall

```
[root@csa-client sysconfig]# iptables --list
Chain INPUT (policy ACCEPT)
target     prot opt source                destination           state RELATED,ESTABLISHED
ACCEPT     all  --  anywhere              anywhere
ACCEPT     icmp --  anywhere              anywhere
ACCEPT     all  --  anywhere              anywhere
ACCEPT     tcp  --  anywhere              anywhere              state NEW tcp dpt:ssh
ACCEPT     tcp  --  anywhere              anywhere              state NEW tcp dpt:5666
REJECT     all  --  anywhere              anywhere              reject-with icmp-host-prohibited

Chain FORWARD (policy ACCEPT)
target     prot opt source                destination           reject-with icmp-host-prohibited
REJECT     all  --  anywhere              anywhere

Chain OUTPUT (policy ACCEPT)
target     prot opt source                destination
```

Figure1.2 40 List of iptables entries

```
CentOS Linux 7 (Core)
Kernel 3.10.0-1160.114.2.el7.x86_64 on an x86_64

csa-server login: root
Password:
Last login: Wed Apr 24 05:26:40 on tty1
[root@csa-server ~]# cd /usr/local/nagios/etc
[root@csa-server etc]# ls
cgi.cfg  htpasswd.users  nagios.cfg  objects  resource.cfg
[root@csa-server etc]# _
```

Figure1.2 41 Contents of directory containing Nagios server configuration file

```
#####
#
# NAGIOS.CFG - Sample Main Config File for Nagios 4.5.1
#
# Read the documentation for more information on this configuration
# file. I've provided some comments here, but things may not be so
# clear without further explanation.
#
#
#####

# LOG FILE
# This is the main log file where service and host events are logged
# for historical purposes. This should be the first option specified
# in the config file!!!

log_file=/usr/local/nagios/var/nagios.log

# OBJECT CONFIGURATION FILE(S)
# These are the object configuration files in which you define hosts,
# host groups, contacts, contact groups, services, etc.
# You can split your object definitions across several config files
# if you wish (as shown below), or keep them all in a single config file.

# You can specify individual object config files as shown below:
cfg_file=/usr/local/nagios/etc/objects/commands.cfg
cfg_file=/usr/local/nagios/etc/objects/contacts.cfg
cfg_file=/usr/local/nagios/etc/objects/timeperiods.cfg
cfg_file=/usr/local/nagios/etc/objects/templates.cfg

# Definitions for monitoring the local (Linux) host
cfg_file=/usr/local/nagios/etc/objects/localhost.cfg

:q!_
```

Figure1.2 42 Content of 'nagios.cfg' file

```
[root@csa-server etc]# cd objects
[root@csa-server objects]# ls
commands.cfg  localhost.cfg  switch.cfg    timeperiods.cfg
contacts.cfg  printer.cfg    templates.cfg windows.cfg
[root@csa-server objects]# _
```

Figure1.2 43 Contents of 'objects' folder

```
[root@csa-server objects]# cp localhost.cfg linuxclient.cfg
[root@csa-server objects]# vi linuxclient.cfg_
```

Figure1.2 44 Copy and edit Linux template for Nagios client

```

#
# NOTE: This config file is intended to serve as an *extremely* simple
#       example of how you can create configuration entries to monitor
#       the local (Linux) machine.
#
#####

#####
#
# HOST DEFINITION
#
#####

# Define a host for the local machine

define host {

    use                linux-server                ; Name of host template to use
                ; This host definition will inherit all variable
s that are defined                ; in (or inherited by) the linux-server host tem

plate definition.
    host_name        csa-client
    alias             csa-client
    address           10.0.1.21
}

#####

```

Figure1.2 45 Modify the host definition

```

# Define a service to "ping" the local machine

define service {

    use                generic-service          ; Name of service template to use
    host_name          csa-client
    service_description PING
    check_command       check_ping!100.0,20%!500.0,60%
}

# Define a service to check the disk space of the root partition
# on the local machine. Warning if < 20% free, critical if
# < 10% free space on partition.

define service {

    use                generic-service          ; Name of service template to use
    host_name          csa-client
    service_description Root Partition
    check_command       check_local_disk!20%!10%!/
}

# Define a service to check the number of currently logged in
# users on the local machine. Warning if > 20 users, critical
# if > 50 users.

define service {

    use                generic-service          ; Name of service template to use
    host_name          csa-client
    service_description Current Users

```

Figure1.2 46 Modify hostname for service definitions

```

define service {
    use                generic-service        ; Name of service template to use
    host_name          csa-client
    service_description Current Users
    check_command       check_local_users!20!50
}

# Define a service to check the number of currently running procs
# on the local machine. Warning if > 250 processes, critical if
# > 400 processes.

define service {
    use                generic-service        ; Name of service template to use
    host_name          csa-client
    service_description Total Processes
    check_command       check_local_procs!250!400!RSZDT
}

# Define a service to check the load on the local machine.

define service {
    use                generic-service        ; Name of service template to use
    host_name          csa-client
    service_description Current Load
    check_command       check_local_load!5.0,4.0,3.0!10.0,6.0,4.0
}

-

```

Figure1.2 47 Hostname for service definitions modified

```

[root@csa-server objects]# ls
commands.cfg  linuxclient.cfg  printer.cfg  templates.cfg  windows.cfg
contacts.cfg  localhost.cfg    switch.cfg   timeperiods.cfg
[root@csa-server objects]# ls -l
total 60
-rw-rw-r--. 1 nagios nagios 6757 Apr 23 10:13 commands.cfg
-rw-rw-r--. 1 nagios nagios 1803 Apr 24 05:00 contacts.cfg
-rw-r--r--. 1 root  root   4461 Apr 25 10:43 linuxclient.cfg
-rw-rw-r--. 1 nagios nagios 4777 Apr 23 10:13 localhost.cfg
-rw-rw-r--. 1 nagios nagios 3001 Apr 23 10:13 printer.cfg
-rw-rw-r--. 1 nagios nagios 3484 Apr 23 10:13 switch.cfg
-rw-rw-r--. 1 nagios nagios 12533 Apr 23 10:13 templates.cfg
-rw-rw-r--. 1 nagios nagios 3512 Apr 23 10:13 timeperiods.cfg
-rw-rw-r--. 1 nagios nagios 4074 Apr 23 10:13 windows.cfg
[root@csa-server objects]# _

```

Figure1.2 48 Ownership of files in ‘objects’ folder

```
[root@csa-server objects]# chown nagios:nagios linuxclient.cfg
[root@csa-server objects]# ls -l
total 60
-rw-rw-r--. 1 nagios nagios 6757 Apr 23 10:13 commands.cfg
-rw-rw-r--. 1 nagios nagios 1803 Apr 24 05:00 contacts.cfg
-rw-r--r--. 1 nagios nagios 4461 Apr 25 10:43 linuxclient.cfg
-rw-rw-r--. 1 nagios nagios 4777 Apr 23 10:13 localhost.cfg
-rw-rw-r--. 1 nagios nagios 3001 Apr 23 10:13 printer.cfg
-rw-rw-r--. 1 nagios nagios 3484 Apr 23 10:13 switch.cfg
-rw-rw-r--. 1 nagios nagios 12533 Apr 23 10:13 templates.cfg
-rw-rw-r--. 1 nagios nagios 3512 Apr 23 10:13 timeperiods.cfg
-rw-rw-r--. 1 nagios nagios 4074 Apr 23 10:13 windows.cfg
[root@csa-server objects]#
```

Figure1.2 49 Change ownership of Nagios client configuration file

```
[root@csa-server etc]# cd ..
[root@csa-server nagios]# vi nagios.cfg
```

Figure1.2 50 Add Nagios client to 'nagios.cfg' file

```
#####
#
# MAGI0S.CFG - Sample Main Config File for Nagios 4.5.1
#
# Read the documentation for more information on this configuration
# file. I've provided some comments here, but things may not be so
# clear without further explanation.
#
#
#####

# LOG FILE
# This is the main log file where service and host events are logged
# for historical purposes. This should be the first option specified
# in the config file!!!

log_file=/usr/local/nagios/var/nagios.log

# OBJECT CONFIGURATION FILE(S)
# These are the object configuration files in which you define hosts,
# host groups, contacts, contact groups, services, etc.
# You can split your object definitions across several config files
# if you wish (as shown below), or keep them all in a single config file.

# You can specify individual object config files as shown below:
cfg_file=/usr/local/nagios/etc/objects/commands.cfg
cfg_file=/usr/local/nagios/etc/objects/contacts.cfg
cfg_file=/usr/local/nagios/etc/objects/timeperiods.cfg
cfg_file=/usr/local/nagios/etc/objects/templates.cfg

# Definitions for monitoring the local (Linux) host
cfg_file=/usr/local/nagios/etc/objects/localhost.cfg
cfg_file=/usr/local/nagios/etc/objects/linuxclient.cfg
"nagios.cfg" 137BL, 45897C
```

Figure1.2 51 Nagios client added

Nagios: 10.0.1.2

Current Network Status
Last Updated: Fri Apr 26 13:28:56 EDT 2024
Updated every 90 seconds
Nagios® Core™ 4.5.1 - www.nagios.org
Logged in as nagiosadmin

Host Status Totals
Up: 2, Down: 0, Unreachable: 0, Pending: 0
All Problems: 0, All Types: 2

Service Status Totals
OK: 15, Warning: 0, Unknown: 0, Critical: 2, Pending: 0
All Problems: 2, All Types: 16

Service Status Details For All Hosts
Limit Results: 100

Host	Service	Status	Last Check	Duration	Attempt	Status Information
csa-client	Current Load	OK	04-26-2024 13:24:42	1d 2h 4m 6s	1/3	OK - load average: 0.39, 0.18, 0.06
	Current Users	OK	04-26-2024 13:25:38	1d 2h 2m 51s	1/3	USERS OK - 1 users currently logged in
	HTTP	CRITICAL	04-26-2024 13:26:34	1d 1h 57m 36s	3/3	connect to address 10.0.1.3 and port 80: No route to host
	PING	OK	04-26-2024 13:27:31	0d 0h 1m 25s	1/3	PING OK - Packet loss = 0%, RTA = 0.70 ms
	Root Partition	OK	04-26-2024 13:28:27	1d 1h 59m 6s	1/3	DISK OK - free space: / 15879 MB (88.82% inode=100%)
	SSH	CRITICAL	04-25-2024 12:59:48	1d 1h 19m 8s	3/3	connect to address 10.0.1.3 and port 22: No route to host
	Swap Usage	OK	04-25-2024 13:00:44	1d 1h 56m 36s	1/3	SWAP OK - 100% free (2047 MB out of 2047 MB)
	Total Processes	OK	04-25-2024 13:01:41	1d 1h 55m 21s	1/3	PROCS OK: 34 processes with STATE = RSZDT
localhost	Current Load	OK	04-26-2024 13:25:10	1d 1h 23m 21s	1/4	OK - load average: 0.23, 0.16, 0.06
	Current Users	OK	04-26-2024 13:26:06	1d 1h 22m 25s	1/4	USERS OK - 1 users currently logged in
	HTTP	OK	04-26-2024 13:27:02	1d 1h 21m 29s	1/4	HTTP OK: HTTP/1.1 200 OK - 303 bytes in 0.010 second response time
	PING	OK	04-26-2024 13:27:59	1d 1h 20m 32s	1/4	PING OK - Packet loss = 0%, RTA = 0.17 ms
	Root Partition	OK	04-26-2024 13:28:13	1d 1h 22m 45s	1/4	DISK OK - free space: / 15879 MB (88.82% inode=100%)
	SSH	OK	04-26-2024 13:26:25	1d 1h 20m 57s	1/4	SSH OK - OpenSSH_7.4 (protocol 2.0)
	Swap Usage	OK	04-26-2024 13:28:38	1d 1h 23m 51s	1/4	SWAP OK - 100% free (2047 MB out of 2047 MB)
	Total Processes	OK	04-26-2024 13:25:58	1d 1h 24m 35s	1/4	PROCS OK: 38 processes with STATE = RSZDT

Results 1 - 16 of 16 Matching Services

Waiting for www.youtube.com...

Page Tour

Figure1.2 52 Service status details for Nagios server and client

2 DNS Caching Setup with CentOS Server and Fedora Clients

2.1 Configuring CentOS Server for DNS Caching

```
[root@csa-server ~]# yum install bind bind-utils
Loaded plugins: fastestmirror
Loading mirror speeds from cached hostfile
 * base: bd.mirror.vanhost.com
 * extras: centos.excellmedia.net
 * updates: bd.mirror.vanhost.com
Package 32:bind-9.11.4-26.P2.el7_9.15.x86_64 already installed and latest version
Package 32:bind-utils-9.11.4-26.P2.el7_9.15.x86_64 already installed and latest version
Nothing to do
[root@csa-server ~]# _
```

Figure 2.1 1 Install packages to provide named service and tools to manage it

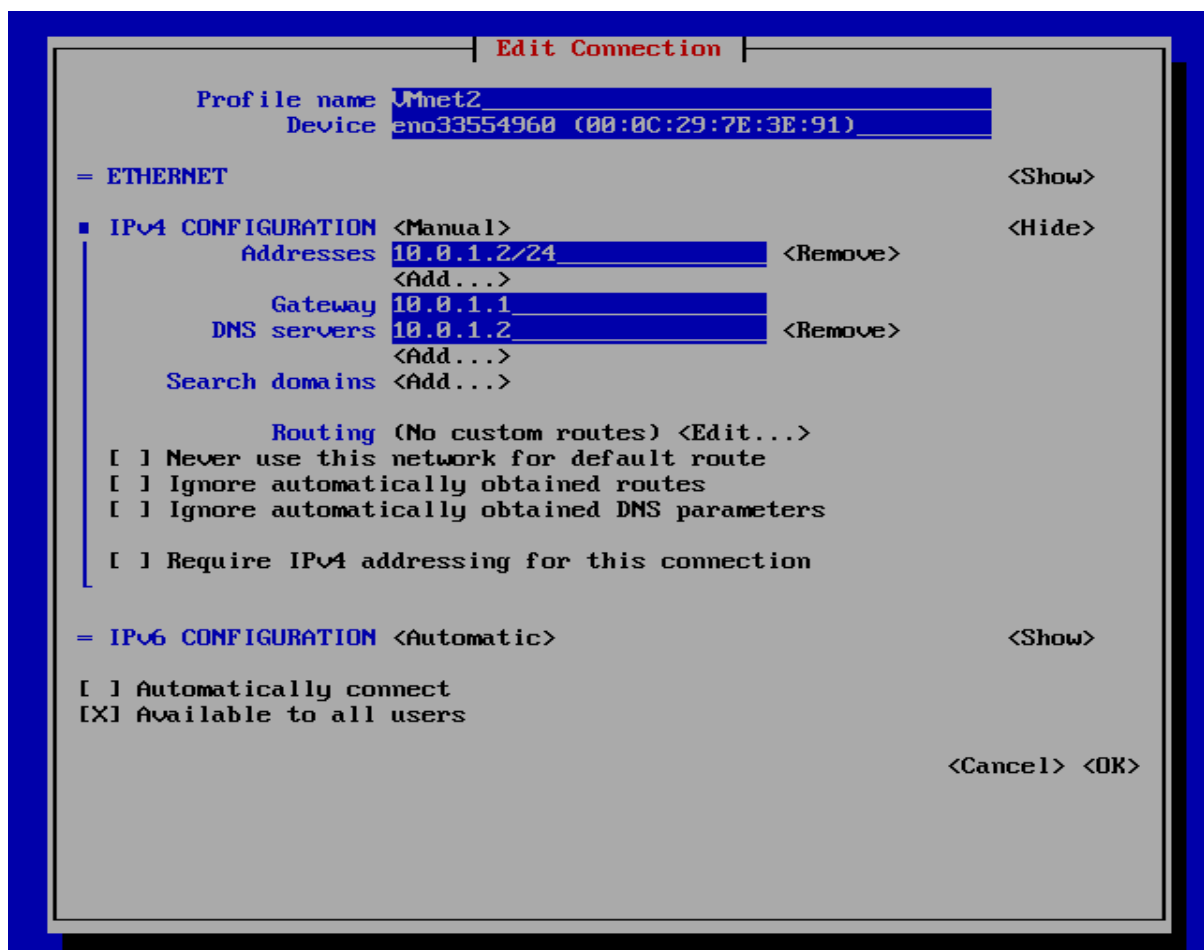


Figure 2.1 2 Caching name server configured

```
[root@csa-server ~]# vi /etc/resolv.conf_
```

Figure 2.1 3 Edit the ‘/etc/resolv.conf’ file

```
# Generated by NetworkManager
nameserver 10.0.1.2_

-- INSERT --
```

Figure 2.1 4 ‘/etc/resolv.conf’ file edited

```
[root@csa-server ~]# vi /etc/named.conf_
```

Figure 2.1 5 Edit DNS's main configuration file

```
//
// named.conf
//
// Provided by Red Hat bind package to configure the ISC BIND named(8) DNS
// server as a caching only nameserver (as a localhost DNS resolver only).
//
// See /usr/share/doc/bind*/sample/ for example named configuration files.
//
// See the BIND Administrator's Reference Manual (ARM) for details about the
// configuration located in /usr/share/doc/bind-{version}/Bv9ARM.html

options {
    listen-on port 53 { 127.0.0.1; 10.0.1.2; };
    listen-on-v6 port 53 { ::1; };
    directory "/var/named";
    dump-file "/var/named/data/cache_dump.db";
    statistics-file "/var/named/data/named_stats.txt";
    memstatistics-file "/var/named/data/named_mem_stats.txt";
    recursing-file "/var/named/data/named.recursing";
    secroots-file "/var/named/data/named.secroots";
    allow-query { localhost; any; };

    /*
    - If you are building an AUTHORITATIVE DNS server, do NOT enable recursion.
    - If you are building a RECURSIVE (caching) DNS server, you need to enable
      recursion.
    - If your recursive DNS server has a public IP address, you MUST enable access
      control to limit queries to your legitimate users. Failing to do so will
      cause your server to become part of large scale DNS amplification
      attacks. Implementing BCP38 within your network would greatly
      reduce such attack surface
    */
    recursion yes;

    dnssec-enable yes;
    dnssec-validation yes;
}

"/etc/named.conf" 75L, 2051C
```

Figure 2.1 6 DNS's main configuration file edited

```
[root@csa-server ~]# systemctl restart named
[root@csa-server ~]#
```

Figure 2.1 7 Restart the named service

```
[root@csa-server ~]# systemctl restart named
[root@csa-server ~]# netstat -antp
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 127.0.0.1:53            0.0.0.0:*               LISTEN      1533/named
tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN      948/sshd
tcp        0      0 127.0.0.1:953           0.0.0.0:*               LISTEN      1533/named
tcp        0      0 127.0.0.1:25            0.0.0.0:*               LISTEN      1165/master
tcp6       0      0 :::80                   :::*                    LISTEN      951/httpd
tcp6       0      0 :::1:53                  :::*                    LISTEN      1533/named
tcp6       0      0 :::22                    :::*                    LISTEN      948/sshd
tcp6       0      0 :::1:953                  :::*                    LISTEN      1533/named
tcp6       0      0 :::1:25                   :::*                    LISTEN      1165/master
tcp6       0      0 :::443                   :::*                    LISTEN      951/httpd
[root@csa-server ~]# _
```

Figure 2.1 8 List all opened TCP ports

```

[root@csa-server ~]# netstat -antu
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 127.0.0.1:53            0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.1:953           0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.1:25            0.0.0.0:*               LISTEN
tcp6       0      0 :::80                   :::*                     LISTEN
tcp6       0      0 :::1:53                  :::*                     LISTEN
tcp6       0      0 :::22                     :::*                     LISTEN
tcp6       0      0 :::1:953                  :::*                     LISTEN
tcp6       0      0 :::1:25                    :::*                     LISTEN
tcp6       0      0 :::443                     :::*                     LISTEN
tcp6       0      0 127.0.0.1:43258         127.0.0.1:43258        TIME_WAIT
udp        0      0 127.0.0.1:53            0.0.0.0:*               LISTEN
udp6       0      0 :::1:53                   :::*                     LISTEN
[root@csa-server ~]#

```

Figure 2.1 9 List all opened ports

```

[root@csa-server ~]# dig google.com_

```

Figure 2.1 10 Display answer from ‘google.com’

```

;; ADDITIONAL SECTION:
e.gtld-servers.net. 172800 IN A 192.12.94.30
b.gtld-servers.net. 172800 IN A 192.33.14.30
a.gtld-servers.net. 172800 IN A 192.5.6.30
i.gtld-servers.net. 172800 IN A 192.31.80.30
f.gtld-servers.net. 172800 IN A 192.35.51.30
j.gtld-servers.net. 172800 IN A 192.48.79.30
k.gtld-servers.net. 172800 IN A 192.52.178.30
c.gtld-servers.net. 172800 IN A 192.26.92.30
g.gtld-servers.net. 172800 IN A 192.42.93.30
h.gtld-servers.net. 172800 IN A 192.54.112.30
l.gtld-servers.net. 172800 IN A 192.41.162.30
m.gtld-servers.net. 172800 IN A 192.55.83.30
e.gtld-servers.net. 172800 IN AAAA 2001:502:1ca1::30
b.gtld-servers.net. 172800 IN AAAA 2001:503:231d::2:30
a.gtld-servers.net. 172800 IN AAAA 2001:503:a83e::2:30
d.gtld-servers.net. 172800 IN AAAA 2001:500:856e::30
i.gtld-servers.net. 172800 IN AAAA 2001:503:39c1::30
f.gtld-servers.net. 172800 IN AAAA 2001:503:d414::30
j.gtld-servers.net. 172800 IN AAAA 2001:502:7094::30
k.gtld-servers.net. 172800 IN AAAA 2001:503:d2d::30
c.gtld-servers.net. 172800 IN AAAA 2001:503:83eb::30
g.gtld-servers.net. 172800 IN AAAA 2001:503:eea3::30
h.gtld-servers.net. 172800 IN AAAA 2001:502:8cc::30
l.gtld-servers.net. 172800 IN AAAA 2001:500:d937::30
m.gtld-servers.net. 172800 IN AAAA 2001:501:b1f9::30

;; Query time: 1277 msec
;; SERVER: 10.0.1.2#53(10.0.1.2)
;; WHEN: Fri Apr 26 10:13:36 EDT 2024
;; MSG SIZE rcvd: 851

```

Figure 2.1 11 Display answer from ‘google.com’ again

```

com. 172770 IN NS m.gtld-servers.net.
com. 172770 IN NS k.gtld-servers.net.

;; ADDITIONAL SECTION:
e.gtld-servers.net. 172770 IN A 192.12.94.30
b.gtld-servers.net. 172770 IN A 192.33.14.30
a.gtld-servers.net. 172770 IN A 192.5.6.30
d.gtld-servers.net. 172770 IN A 192.31.80.30
i.gtld-servers.net. 172770 IN A 192.43.172.30
f.gtld-servers.net. 172770 IN A 192.35.51.30
j.gtld-servers.net. 172770 IN A 192.48.79.30
k.gtld-servers.net. 172770 IN A 192.52.178.30
c.gtld-servers.net. 172770 IN A 192.26.92.30
g.gtld-servers.net. 172770 IN A 192.42.93.30
h.gtld-servers.net. 172770 IN A 192.54.112.30
l.gtld-servers.net. 172770 IN A 192.41.162.30
m.gtld-servers.net. 172770 IN A 192.55.83.30
e.gtld-servers.net. 172770 IN AAAA 2001:502:1ca1::30
b.gtld-servers.net. 172770 IN AAAA 2001:503:231d::2:30
a.gtld-servers.net. 172770 IN AAAA 2001:503:a83e::2:30
d.gtld-servers.net. 172770 IN AAAA 2001:500:856e::30
i.gtld-servers.net. 172770 IN AAAA 2001:503:39c1::30
f.gtld-servers.net. 172770 IN AAAA 2001:503:d414::30
j.gtld-servers.net. 172770 IN AAAA 2001:502:7094::30
k.gtld-servers.net. 172770 IN AAAA 2001:503:d2d::30
c.gtld-servers.net. 172770 IN AAAA 2001:503:83eb::30
g.gtld-servers.net. 172770 IN AAAA 2001:503:eea3::30
h.gtld-servers.net. 172770 IN AAAA 2001:502:8cc::30
l.gtld-servers.net. 172770 IN AAAA 2001:500:d937::30
m.gtld-servers.net. 172770 IN AAAA 2001:501:b1f9::30

;; Query time: 0 msec
;; SERVER: 10.0.1.2#53(10.0.1.2)
;; WHEN: Fri Apr 26 10:14:06 EDT 2024
;; MSG SIZE rcvd: 851

```

Figure 2.1 12 Caching name server verified

```

[root@csa-server data]# rndc dumpdb -cache
[root@csa-server data]# cd /var/named/data
[root@csa-server data]# ls -la
total 228
drwxrwx---. 2 named named 92 Apr 26 18:21 .
drwxrwx---T. 5 root named 4096 Apr 25 14:58 ..
-rw-r--r--. 1 named named 583 Apr 26 18:22 cache_dump.db
-rw-r--r--. 1 named named 121492 Apr 26 18:22 named.run
-rw-r--r--. 1 named named 2017 Mar 15 09:18 named.run-20240418
-rw-r--r--. 1 named named 90792 Apr 20 10:10 named.run-20240425

google.com. 171225 NS ns2.google.com.
171225 NS ns1.google.com.
171225 NS ns3.google.com.
171225 NS ns4.google.com.
ns1.google.com. 171225 A 216.239.32.10
ns2.google.com. 171225 A 216.239.34.10
ns3.google.com. 171225 A 216.239.36.10
ns4.google.com. 171225 A 216.239.38.10

```

Figure 2.1 13 View cached data

```
[root@csa-server data]# firewall-cmd --zone=public --add-service=dns
success
[root@csa-server data]# firewall-cmd --get-default-zone
public
```

Figure 2.1 14 Allow DNS traffic through firewall

2.2 Configuring Fedora Client to Utilize CentOS DNS Caching Server

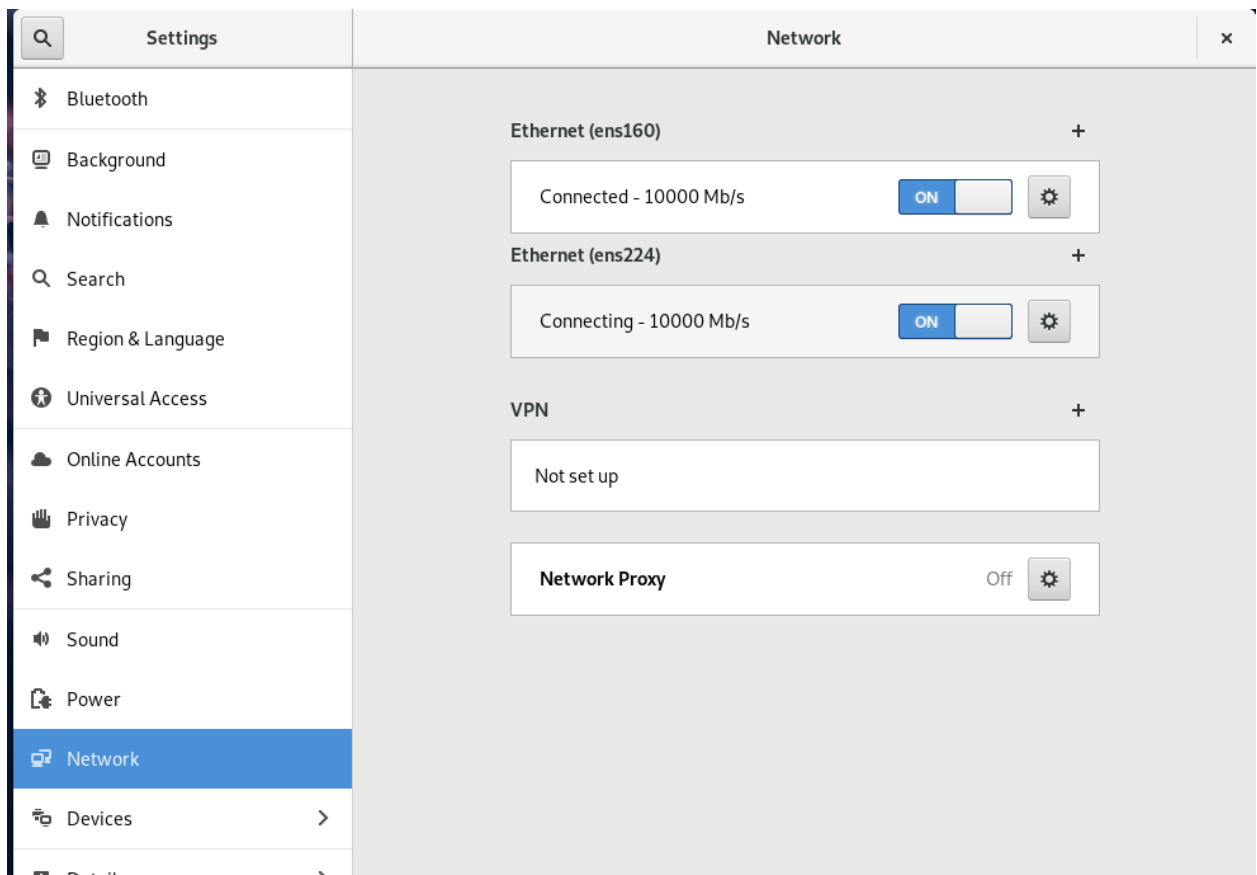


Figure 2.2 1 up connection

```
[sawera@csa-client ~]$ sudo vi /etc/resolv.conf
```

Figure 2.2.2 Edit the '/etc/resolv.conf' file

[illegible]

Figure 2.2 3 ‘/etc/resolv.conf’ file edited

2.3 Conducting DNS Resolution Tests on Fedora Client

```
root@csa-client:~# dig google.com
; <<>> DiG 9.11.10-RedHat-9.11.10-1.fc29 <<>> google.com
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: SERVFAIL, id: 35666
;; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4096
; COOKIE: f52ee116abc67260de29cf88662a8f3d847155fd88cedc39 (good)
;; QUESTION SECTION:
;google.com.                                IN      A

;; Query time: 13 msec
;; SERVER: 10.0.1.2#53(10.0.1.2)
;; WHEN: Thu Apr 25 07:43:32 EDT 2024
;; MSG SIZE rcvd: 67
```

Figure 2.3 1 Resolution test for ‘google.com’

References

[1]

“Nagios Core Part 1: Server (CentOS) and Client (Linux & Windows) Setup Tutorial,” www.youtube.com.

https://www.youtube.com/watch?v=BtaSY1ALZzE&list=PLKAyQSSPCBRp0zIcbfVw1Jd33T4QxOu_v&ab_channel=SysAdmGirl (accessed Apr. 14, 2024).

[1] Manoj, "How to Install and Configure Nagios on CentOS 7?," Aalam Info Solutions LL, 14 Feb 2023. [Online]. Available: <https://medium.com/@aalam-info-solutions-llp/how-to-install-and-configure-nagios-on-centos-7-befcbf04ca33>. [Accessed 20 Apr 2024].